

Differential Reflectance Spectroscopy for Real-Time Reflectance measurement

Department of Physics, Freie Universität Berlin

Supervisor: Dr. Cornelius Gahl

Md Ramjan Ali

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1 Abstract

Among various spectroscopic methods, Dependent Differential Reflectance Spectroscopy (*pol – DRS*) is a powerful, non-destructive technique which is used for real time, **In-Situ** monitoring of organic thin film growth with mono or sub-monolayer resolutions. Differential Reflectance Spectroscopy (*pol – DRS*) continues to monitor the changes of the sample's reflectance upon deposition or modification, providing information about the evolution of optical properties which in turn gives information about the grown film. This **extended pol-DRS** technique measures the reflectance changes for both **-S and -P** polarization states of the incident light at oblique incidence. The concomitant measurement of the polarization-dependent DRS signals helps to determine the growing film's optical anisotropy and immediately linked to the molecular alignment; structural formation on the substrate surface which is crucial for understanding fundamental growth mechanisms including nucleation, layer formation and the transition between different growth modes that represents the electronic and functional properties of the final organic thin layer. By deploying **pol-DRS**, insights are gained from the early stages of film formation and the preferred orientation of the molecules, such as the growth of perfluoropentacene thin films on Ag(110) substrate. This methodology offers a very wide and significant advantages over any other process in optimizing organic thin film deposition processes for applications in organic electronics [3].

2 Introduction

The ongoing rapid development of high performance organic electronic devices including Light-Emitting Diodes (OLEDs), Field-Effect Transistors (OFETs), Solar cells entirely depends on precise control over the film's morphology, structural and molecular orientation [4] of the constituent. The ultimate functional properties of these devices are determined by process of molecular organization achieved during the entire growth stage that often entailed complex interplay of the arriving organic molecules and the substrate surface. Therefore, in situ real-time monitoring techniques are essential to understand the fundamental processes of thin film nucleation and growth, allowing optimization of deposition parameters to achieve the desired device characteristics [3]. Various Traditional techniques like Atomic Force Microscopy (AFM) and X-ray Diffraction (XRD) comes with a broad structural information but are often ex situ (requiring removal from the growth chamber) or slow, often failing to

trace the dynamic and non-equilibrium processes that govern the initial sub-monolayer growth kinetics. Because of having experimental shortcoming, we must have something that adopt this problem by facilitating updated and fast, non-destructive optical methods that functions effectively, getting along with Ultra-High Vacuum (UHV) or inert atmosphere environments organic thin film deposition techniques like Organic Molecular Beam Epitaxy (MBE), organic molecular beam deposition.

This paper,"Polarization-dependent differential reflectance spectroscopy", comes with a novel way of real-time monitoring of **In-Situ** organic (/inorganic; out of scope of this paper) thin-film growth on solid surface by presenting an extended [3] Differential Reflectance Spectroscopy (DRS) setup that monitors both linear polarization states of reflected light (s-polarized and p-polarized) which is called polarization-dependent DRS (pol-DRS). The main objective of this work is to demonstrate the pol-DRS as a powerful, non-destructive tool for real-time monitoring of thin film growth by measuring the changes of reflectance ($\frac{\Delta R}{R}$) intensity (**I**) as a function of wavelength (λ). By extending this method to incorporate polarization dependence method monitoring the reflectance change for both the *S*- and *P*-polarized light at non-normal incidence, pol-DRS becomes uniquely sensitive to the optical anisotropy of the growing film [4]. This anisotropy is a direct spectroscopic signature of the molecular tilt, alignment and long-range structural order within the organic layer, making pol-DRS an invaluable tool for characterizing the early stages of organic thin film assembly in real-time.

3 Fundamental Physics and Theory

3.1 Properties of Light

Light can be defined broadly by a traveling wave consisting of transverse electric and magnetic field oscillating perpendicular each other and in terms of direction of propagation. Two most important terms that are used to characterize the light wave is wavelength and frequency where the wavelength of light describes the distance between the Crest or trough, while the later parameter describes the round-trip oscillation from top to bottom and back to top number of times per second that a single point can make which is shown in figure-1.

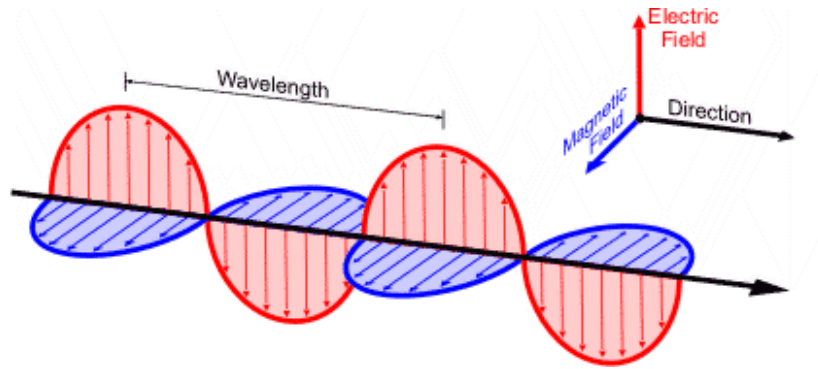


Figure 1: Transverse Electro-Magnetic Wave

3.2 Law of Reflection, Refraction and Absorption

The laws ruling Reflection, Refraction and Absorption are fundamental principles in optics and spectroscopy that refers to the amount of light gets reflected (**r**), refracted (**t**) and absorbed (**a**) when it hits on a surface. According to the conservation of light energy; when light interacts with a surface, either it gets reflected, transmitted (refracted) or absorbed and the total amount of light that hits the surface add up to 100% which is, $r + t + a = 1$.

3.2.1 Reflection

The law of reflection describes the behavior of light bounce off an interface or from a smooth surface, remaining in the same plane after it's get reflected [8]. According to the principle, the angle of incidence θ_i is equal to the angle of reflection (θ_r and incident ray, the reflected ray and the normal (a line perpendicular to the surface at where the light gets hit or contact) all remain in the same plane. So, $\theta_i = \theta_r$ that is shown in figure-2. Reflection can be broadly classified into specular and diffuse reflection which is absolutely depend on the surface's smoothness.

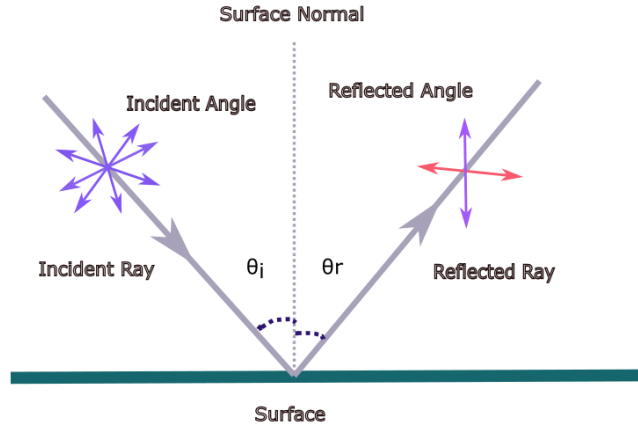


Figure 2: Reflection at Mirror [8]

Any types of smooth surface or mirror produces this type of reflection and the produced image is clear and much visible due to the fact that the angle of incidence and reflection $\theta_i = \theta_r$ is always same. On the contrary, the angle of incidence and reflection ($\theta_i \neq \theta_r$) is not equal for non-smooth surface (as most natural surfaces are rough) and get reflected in many different direction which is called diffuse reflection that is presented in figure-3.

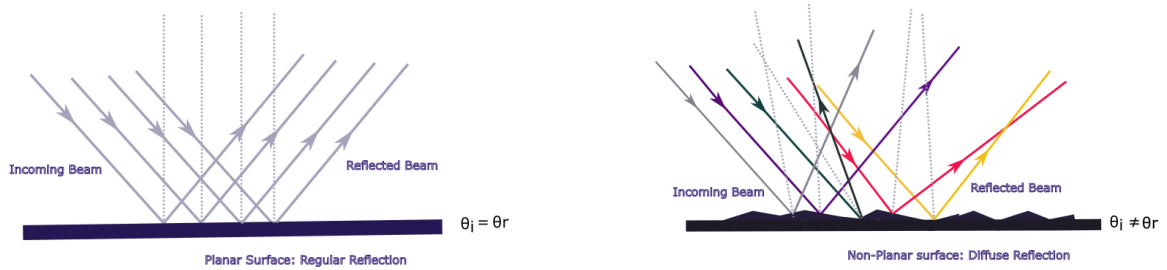


Figure 3: Regular and Specular Reflection [8]

Another notable consequence associated with the law of refraction is when light roving into a medium with a lower index of refraction; light rays bent away from the normal and propagates through the interface between the media. For an incidence angle at which the refracted rays make an angle of 90 with the surface normal is called critical angle θ_c . For incidence larger than the critical angle, the light rays are completely reflected back into the material and this phenomenon is called total internal reflection which can be seen in figure-4.

3.2.2 Law of Refraction [Snell's Law]

The changes in direction or bending of a wave upon it travels from one medium (e.g., from air to water or water to air) to another is called refraction and this effect originates from non-identical density distribution [8] of the two media. The relationship between the angle of incidence along with angle of refraction can be described by using Snell's law,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (1)$$

Snell law describes the ratio [8] of the sine of the angle of incidence (θ_1) to the sine of the angle of refraction (θ_2) is constant for the interface of the two medium,

$$\frac{\sin \theta_i}{\sin \theta_r} = \text{constant}$$

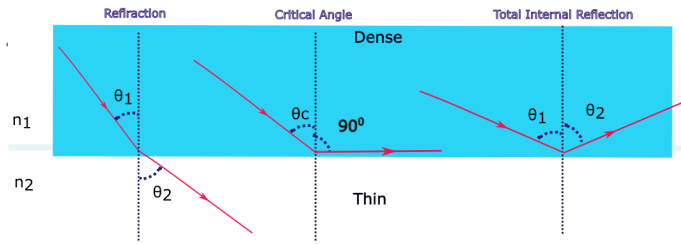


Figure 4: Refraction of Light at Distinct Angle [8]

In above equation (1), n_1 and n_2 are the refractive indices of the first and second medium respectively which intrinsic optical parameter and outlines the relative propagation speed of light in specific medium compared to ones in the vacuum. The relation between optical density and the direction of light path's deviation at an interface is presented in figure-4. The refractive index (phase velocity) can be defined as follows,

$$n = \frac{c}{v} \quad (2)$$

Where n refractive index, c speed of light in vacuum and v speed of light in medium. The higher value of n indicates that this material is more dense and the speed of light is much slower compared to the one in thinner medium which is illustrated in figure-5. From the diagram portrayed in figure-5, it clearly visible that when light ray travels from more optically dense (**Medium 1**) to low dense medium (**Medium 2**), the refracted ray bends away from

surface normal; conversely it bends towards surface normal when it's get refracted from low optically dense (**Medium 1**) to high index medium (**Medium 2**).

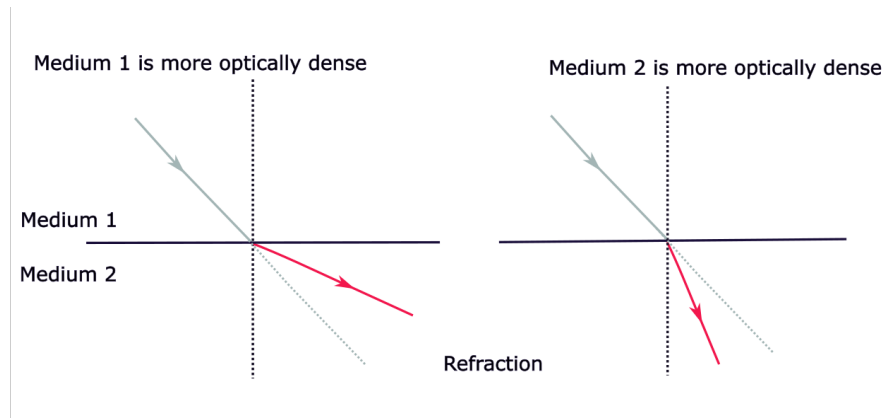


Figure 5: Refraction of Light in the Interface of the two Medium

Recalling refraction which occurs when light passes through a transparent object from another transparent object and bends away from its path and slows down; more prominent in solids [12] than gases or liquids plays a role to understand the unknown properties of any materials by inferring the amount of refraction. The magnitudes of refractive index is always greater than one since light travels fastest in a vacuum with exceptional situations for instance; X-rays propagating through plasmas, or in any engineered meta-materials, the phase velocity can exceed c indexing a refractive index slightly less than one or even negative. The range of refractive indices is usually between 1.5 and 1.7 for glass depending of its type and distinct materials have different refractive indices which is listed in table-1.

3.2.3 Law of Absorption (Beer–Lambert Law)

The absorption of light (photons) is a quantum phenomenon and an energy transfer process where matter (atoms and molecules) absorbs light of specific wavelengths and interacts with it; causing transitions of energy within its discrete energy level (electronic, vibrational and rotational) and which is why objects have color (they reflect what they don't absorb). This event only happens when the light's energy matches the exact difference between energy levels of that atoms or molecules. As individual element or molecule has a unique bond structure, it absorbs at most an unique pattern of wavelengths which creates specific absorption pattern (dark lines) in the spectrum that acts as a fingerprint for that substance. —————need to done—————

Table 1: Index of Refraction (n) for Various Materials

Material	n	Material	n
Vacuum	1.0000	Light Flint Glass	1.58
Air	1.0003	Polycarbonate Glass	1.59
Ice	1.309	Dense Flint Glass	1.66
Water	1.333	Sapphire gemstone	1.77
Ethyl Alcohol	1.361	Zircon	1.923
Teflon	1.38	Diamond	2.417
Fluorite	1.434	Rutile	2.907
Glycerine	1.473	Gallium phosphide	3.50
Benzene	1.501	Plexiglas	1.51

3.3 Polarization in Dielectric Medium

The term, "Polarization" can be defined broadly as the way of how the oscillations of the electric field component ($\vec{E}$) of a electromagnetic-wave restricted to certain plane instead of oscillating in random direction with time. Depending on attenuation of the electric field in certain plane, the polarization has been termed S-polarized (perpendicular) and P-polarized(parallel) light. Any dielectric medium which doesn't host any free charge carriers [49] achieved some degree of polarization inside the material while interacting with imposed electric-field $\vec{E}$ through the collective re-arrangement of the dipoles.

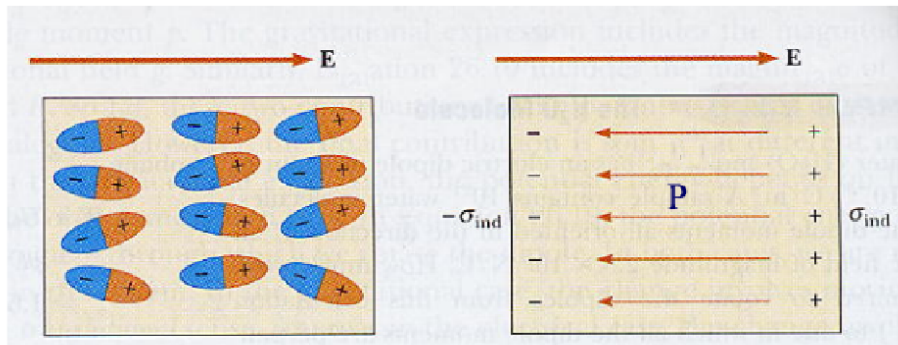


Figure 6: Response of the dielectric medium due to exerted $\vec{E}$ field [49]

Figure 7:

The magnitudes of displacement and polarization due to external field is written by

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = (\epsilon_0 + \chi_E) \vec{E} = \epsilon \vec{E} \quad (3)$$

where, $\chi_E(\vec{E})$ represents the measures of the polarization with material's permittivity ϵ . The subsequent polarization field directed reversely with the applied $\vec{E}$ field and the polarization of that medium can be formulated as follows given that the field intensity is low and the material's composition is same for all direction.

$$\vec{P} = \chi_E(\vec{E}) \vec{E} \quad (4)$$

However, the behavior of the dipole is sort of different when acting field intensity is high; the response of the medium is no longer linear to the applied field which can be seen by the following equation.

$$\vec{P} = \chi_E^{(1)} \vec{E} + \chi_E^{(2)} \vec{E} \vec{E} + \dots \quad (5)$$

In above equation, the first-order term represent linear response and higher terms is non-linear term which give rise nonlinearity.

Fresnel Equations

Fresnel equations are formulas given by French physicist Augustin-Jean Fresnel (1821 – 1823) used to describe the behavior of electromagnetic radiation at the intersection enclosed by two dielectric media having no-identical index of refraction n (medium 1) and n' (medium 2) independently. Fresnel equations precisely describes the magnitude of the reflected and transmitted field of the E-M waves with respect to angle of incidence at the surface. By considering the the electric field propagating [8] in z-direction; the plane wave can be transcribed as,

$$\mathbf{E} = \mathbf{E}_0 e^{i(kz - \omega t)} \quad (6)$$

In above equation, the k represents the waveform of the plane-wave ($k = \frac{2\pi}{\lambda}$) with initial field amplitude E_0 , the angular frequency ω and the K component of the incidence, reflected and transmitted beam is $\vec{k}_i$, $\vec{k}_r$ and $\vec{k}_t$. Since the plane wave is a combination of linearly oscillating $\vec{E}$ and $\vec{M}$ field, the boundary conditions can be applied for each polarization state separately at the interface where it is assumed that the tangential and normal components of the electric $\vec{E}$ and magnetic fields $\vec{B}$ must remain continuous through the boundary in-between

the two media having refractive index n and n' which can be inferred by solving the maxwell equations. The derivation of fresnel equation is manifested as by taking into consideration of the different polarization state of light to sorted out the ratios of reflected and transmitted electric field amplitudes (E_r/E_i and E_t/E_i) with respect to E_i .

S-Polarization [$\vec{E}$ is normal respect to the plane-of incidence]

For S polarization light [figure-10], the polarization of the E field is right angle regards to the interface; according to the boundary condition for transverse electric waves, the tangential component of $\vec{E}$ and the normal components of the $\vec{B}$ of the incidence light can be written by using the trigonometric formula given follow: For incoming beam,

$$\left. \begin{aligned} \cos \theta &= \frac{B_T}{B} \implies B_T = B \cos \theta \\ \sin \theta &= \frac{B_N}{B} \implies B_N = B \sin \theta \end{aligned} \right\} \quad (7)$$

Reflected beam,

$$\left. \begin{aligned} \cos \theta'' &= \frac{B''_T}{B''} \implies B''_T = B'' \cos \theta'' \\ \sin \theta'' &= \frac{B''_N}{B''} \implies B''_N = B'' \sin \theta'' \end{aligned} \right\} \quad (8)$$

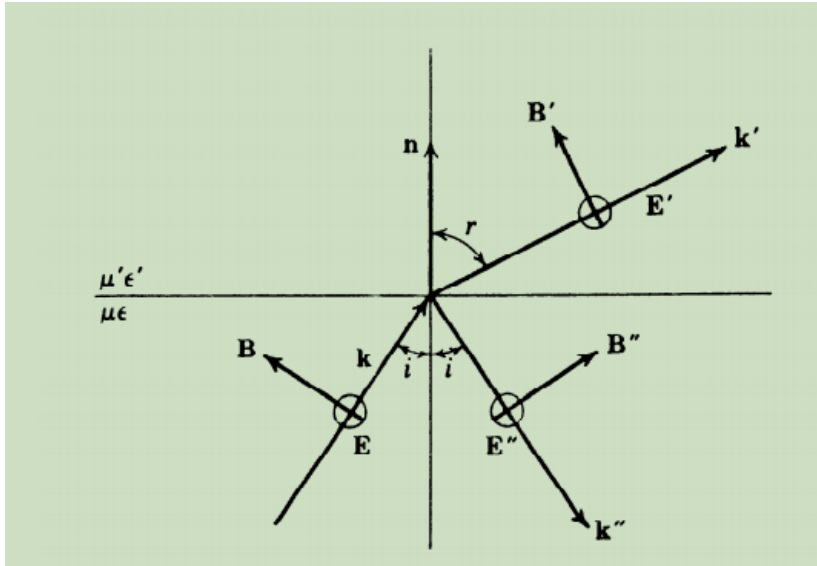


Figure 8: Graphical illustration of Transverse Electric or S-polarization where the incidence($\vec{E}_i$), reflected ($\vec{E}_r$) and transmitted ($\vec{E}_t$) components of E -field are perpendicular respect the plane of incidence (into or out of the screen). [8]

And for transmitted beam,

$$\left. \begin{aligned} \cos \theta' &= \frac{B'_T}{B'} \implies B'_T = B' \cos \theta' \\ \sin \theta' &= \frac{B'_N}{B'} \implies B'_N = B' \sin \theta' \end{aligned} \right\} \quad (9)$$

where the subscripts N, T and i, r, t represents tangential and normal components of the incidence (*i*), reflected (*r*) and transmission (*t*) rays. Since the field must be continuous at the boundary, the tangential component of the $\vec{E}$ and $\vec{B}$ can be written in the form:

$$E + E'' = E' \quad (10)$$

and the components of the magnetic field can be equated as:

$$B \cos \theta - B'' \cos \theta'' = B' \cos \theta' \quad (11)$$

By substituting equation (11) by equations (7)-(9) and considering the relation ($B = \frac{nE}{c}$), the magnetic field components can be expressed as:

$$nE \cos \theta - nE'' \cos \theta'' = n' E' \cos \theta' \quad (12)$$

In the above equation, the angle of reflection has been replaced by angle of incidence and its magnitude are same ($\theta_i = \theta_r$). The equation (12) can be simplified by separating the variable for reflection (E'') and transmission (E') part from equation (10) which in turn gives the ratio of reflection ($r_s = \frac{E''}{E}$) and transmission ($t_s = \frac{E'}{E}$) coefficient respect to the incoming beam.

Reflection coefficient (r_s):

$$r_s = \frac{n \cos \theta - n' \cos \theta'}{n \cos \theta + n' \cos \theta'} \quad (13)$$

Transmission coefficient (t_s)

$$t_s = \frac{2n \cos \theta}{n \cos \theta + n' \cos \theta'} \quad (14)$$

P-Polarization [$\vec{E}$ is parallel respect to the plane of incidence]

For transverse magnetic cases, the oscillating electric field ($\vec{E}$) is parallel or in-plane respect to the plane of incidence which is shown in figure

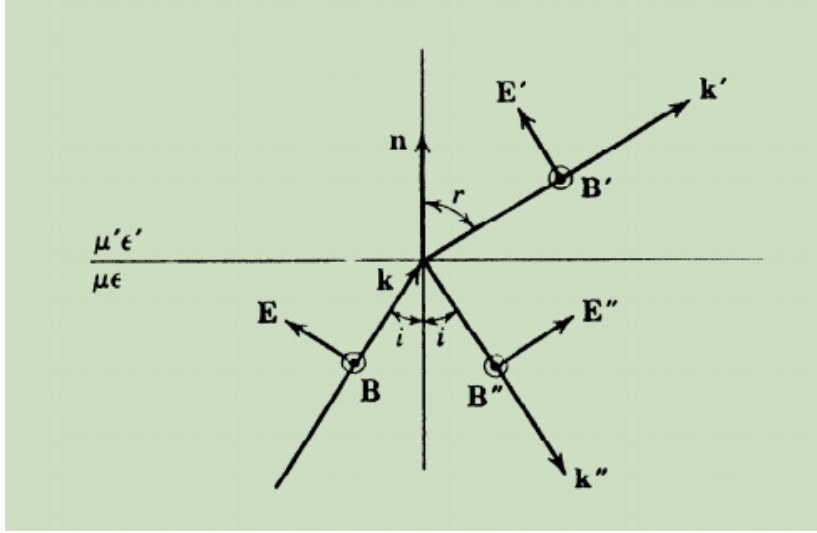


Figure 9: A schematic representation of transverse magnetic or P-polarization projecting the incident ($\vec{E}$), reflected ($\vec{E}''$) and transmitted ($\vec{E}'$) electric field vectors oriented parallel to the plane of incidence. [8]

In terms of transverse-magnetic polarization, the related tangential and the normal components of the ($\vec{E}$) and ($\vec{B}$) incidence, reflected and transmitted beams are expressed as:

$$\left. \begin{aligned} E_T &= E \cos \theta \\ E_N &= E \sin \theta \end{aligned} \right\} \quad (15)$$

$$\left. \begin{aligned} E_T'' &= E'' \cos \theta'' \\ E_N'' &= E'' \sin \theta'' \end{aligned} \right\} \quad (16)$$

$$\left. \begin{aligned} E_T' &= E' \cos \theta' \\ E_N' &= E' \sin \theta' \end{aligned} \right\} \quad (17)$$

Applying boundary conditions for in-plane polarization of $\vec{E}$ field, the magnetic field and electric field can be formulated as follow:

$$B + B'' = B' \quad (18)$$

$$E \cos \theta - E'' \cos \theta'' = E' \cos \theta' \quad (19)$$

Now substituting $B = \frac{n}{c}E$ (where $n = \sqrt{\epsilon\mu}$) and by following the same procedure used in S-polarized beam, the equation (19) can be simplifies and can be written for reflection and

Reflection coefficient (r_p):

Transmission coefficient (t_p):

For normal incidence.....

An Ultra-High Vacuum (UHV) chamber is defined by a sealed metal chamber or enclosure designed to achieve pressures below 10^{-9} mbar (2×10^{-7} Pa). The acronym "UHV" stands for Ultra High Vacuum which is the absence of any types of contaminants inside the chamber or system and requires pressure less than 10^{-7} Torr. The word "vacuum" means empty and it rooted from the Latin word "vacua" [13] which sustains almost empty space in nature or in any system although the fact is that it's quite impossible to have such condition as there are always some sort contaminants remains even after attained vacuum condition. Vacuum is comparatively empty space or volume containing lesser gas particles ("gas" comes from the Greek word "chaos") with a lower pressure and density than neighbouring atmosphere measured residual gas pressure in the chamber. Depending on its configuration, vacuum system can be classified into 3 categories (Rough, high and Ultra high Vacuum) and the units used to express the magnitude are Pa, Torr and mbar ($1 \text{ Pa} \approx 7.5 \times 10^{-3} \text{ Torr}$ and $1 \text{ Pa} = 1 \times 10^{-2} \text{ mbar}$ [13]).

18

vacuum pumps but it also requires appropriate system design, expertise and materials [15] to reduce gettering inside the chamber.

5 Differential Reflectance Spectroscopy

Optical spectroscopy, which is commonly used to monitor physical processes occurring at surfaces and interfaces in real-time, is particularly suitable for studying structural, electronic, and optical properties of molecular thin films before or at the time of deposition, and gives information with sub-monolayer resolution. As light has short mean free path in matter, differential methods like DRS or RDS are used to eliminate the optical response from the bulk and obtain the surface's optical response. This process is as follows:

- Differential Reflectance Spectroscopy (DRS) compares the reflectance of a clean substrate with one that has been modified owing to gas or molecule adsorption on solid where the relative change of the reflectance gives indirect information on the modified surface properties.
- Growth and crystallization on solid surface leads to a strong influence on the optical absorption and make DRS a highly universal method to measure adsorbates on surfaces.
- The DRS signal ($\Delta R/R$) is defined as the change in reflectance $R(E, t)$ after a time t (during deposition) relative to the reflectance of a unmodified or pristine surface at $t = 0$. Mathematically,

$$\begin{aligned} \text{DRS}_{(E,t)} &= \frac{R(E,t) - R(E,0)}{R(E,0)} \\ &= \frac{I(E,t) - I(E,0)}{I(E,0)} \end{aligned} \quad (22)$$

where reflectance R is a function of E and t . The second part of the above equation ($E=I$, Photon-energy=Intensity) comes from the fact that; the reflectance (R) of a surface is directly proportional to the measured intensity (I) of the reflected beam.

- When a film is very thin ($d \ll \lambda$), the real and imaginary parts of the dielectric function can be directly obtained from the DRS measurements.

- For $\theta \approx 0^\circ$, the reflectance from both s -polarized and p -polarized light is identical ($R_s = R_p$) and difference in reflected intensity can only be measured for an optically anisotropic sample. For Oblique Incidence, $\theta = 65^\circ$ the reflectance will depend on the linear polarization even for optically isotropic and anisotropic samples. The DRS produces different signals for s - and p -polarizations, gives information on the electronic states, optical transitions and molecular arrangement of the surface.

The equations for the reflection coefficients for **s-polarized and p-polarized** light are written as;

$$r_p = \frac{N_{21}^2 \cos \theta - \sqrt{N_{21}^2 - \sin^2 \theta}}{N_{21}^2 \cos \theta + \sqrt{N_{21}^2 - \sin^2 \theta}} \quad (23)$$

$$r_s = \frac{\cos \theta - \sqrt{N_{21}^2 - \sin^2 \theta}}{\cos \theta + \sqrt{N_{21}^2 - \sin^2 \theta}} \quad (24)$$

Here, N_{21} is the ratio of the complex refractive index of both materials defined as $N_{21} = N_2/N_1$ and the angle of incidence θ which is measured with respect to the surface normal.

Differential Reflectance Spectroscopy (DRS) measures the reflectance R which is the ratio of the reflected intensity (power) to the incident intensity and since intensity is proportional to the square of the electric field amplitude ($I \propto |E|^2$), the reflectance coefficients are:

$$\mathbf{R}_s = |\mathbf{r}_s|^2 \quad (25)$$

5.1 Kramers–Kronig (KK) Analysis for Complex Permittivity ($|\epsilon|^2$)

The Kramers-Kronig (KK) analysis is an essential tool used in materials science, optics and spectroscopy for studying the complex permittivity $\epsilon = \epsilon_1 + i\epsilon_2$. The spectrum obtained from the typical reflectance ratio measurements only provides information about its magnitude which already has been derived in equation (25) but it doesn't come with any intuitive information about phase [10, 19]. The Kramers-Kronig (KK) method relates the real and imaginary parts of the complex function (such as material's dielectric function or reflection coefficient) which helps for rebuilding and verifying the full complex permittivity from a single measured quantity, such as reflectivity [5, 10, 11, 14].

The very earliest mathematical framework was established solitarily by Ralph Kronig in 1926 and Hendrik Anthony Kramers in 1927 on this. T. S. Robinson was the first to apply K-K analysis to infrared reflectance spectra with the intension of reflectance measurement to determine a sample's index of refraction (n) and extinction coefficient (k) at nearly normal incidence [5, 11, 14]. An extension of this method's applicability was developed by D. M. Roessler in 1965 through introducing algorithm to compute optical constants from absolute reflectance spectra measured at oblique angles ($0^\circ \leq \theta < 90^\circ$) using s-polarized light [5, 11]. Dwight W. Berreman later generalized (1967) this analysis for any incident [5, 14] at almost any angle given incidence (s- or p-polarized radiation). George M. Hale, Wayne E. Holland, and Marvin R. Querry later introduced algorithm 1973 for relative reflectance spectra measurement that compares a sample to a known standard [5, 11]. Further development have been made by Marvin R. Querry and Wayne E. Holland (1974) to analyze reflectance ratio (R_p/R_s) which calculates n and k from the phase shift difference between polarization components [5]. Later in 1991, P. Grosse and V. Offermann modified this method so that it can handle complex layered structures as well as non-perpendicular incidence by introducing normalization procedures and blaschke factors; to account for zeroes in the reflection coefficient [10].

.....need to done

6 Femtosecond Laser Pulse Generation

A femtosecond laser can be characterized by laser system that emits extremely short few cycle optical [16, 23] pulses generally lasting less than one picosecond (ultrashort pulse lasers) and belongs to femtosecond (10^{-15} s) or sub-femtosecond (10^{-15} s) range. These type of pulses are usually produced by using passive mode locking; a way that yields a a rapid, regular sequence of pulses (high repetition rate) extending from megahertz [16] to gigahertz range. However, the average output power of the laser is often limited leading to low pulse energies belongs to the nano-joule nJ range which then need to amplify by using external amplifiers to achieve much higher energies for particular application [18] from testing high-performance semiconductor devices, smooth processing of materials, initiating and tracing chemical reactions in addition to complex surgical applications. There are different era of science which was inaccessible before now feasible due to the application of ultrashort light pulses such as non-equilibrium dynamics and interactions of light with matters. The

sequential steps involves in femtosecond pulse generation is presented as a diagram in figure (11).

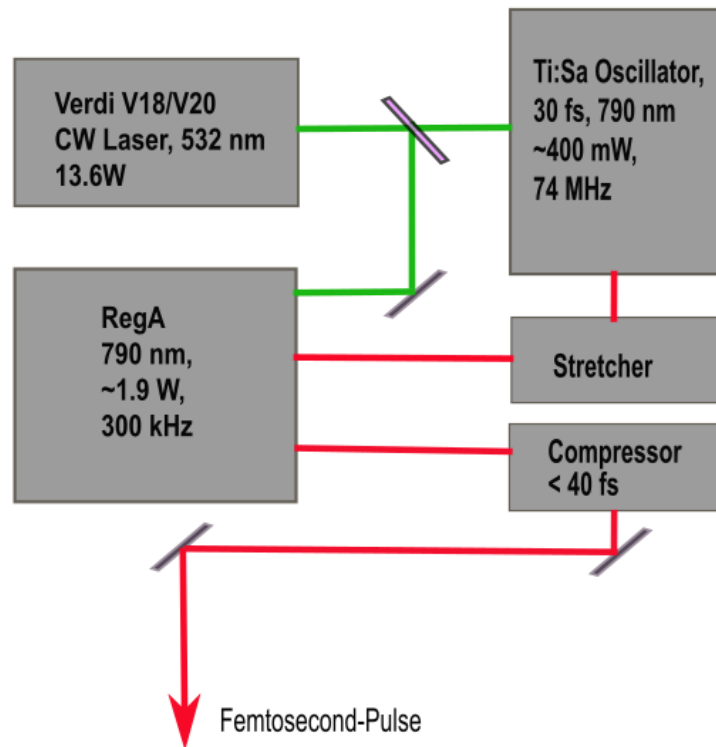


Figure 10: Design of femto-second pulse generation stages including pump laser, oscillator, amplifier (RegA, stretcher), compressor and optical parametric amplifier (OPA) [32,45]

The unique ability of the femtosecond lasers to concentrate significant optical field strength both in the time and space domains enables to induce powerful non-linear light–matter interactions;making vital for advanced technological applications [17].The provided table (2) summarizes ultra-fast dynamics on different time-domain and associated characteristic processes.

Table 2: Ultra-fast dynamics in different timescales and characteristic processes [18]

Timescale	Magnitude	Characteristic Processes
Attosecond (as)	10^{-18} s	Electronic Motion: The fastest events, including the motion of electrons in atoms and molecules, photoionization, and core-level relaxation.
Femtosecond (fs)	10^{-15} s	Molecular Dynamics: Molecular vibrations, bond breaking and formation in chemical reactions, the first steps of vision and photosynthesis, and electron-phonon thermalization in solids.
Picosecond (ps)	10^{-12} s	Energy Transfer and Relaxation: Rotational motion of molecules in liquids, heat diffusion, non-radiative energy transfer, and vibrational relaxation.
Nanosecond (ns)	10^{-9} s	Macromolecular Dynamics: Inter-system crossing, fluorescence lifetimes, and initial steps of protein folding.
Microsecond (μ s)	10^{-6} s	Slower Biological/Chemical Events: Phosphorescence, large conformational changes in proteins, and some enzyme reactions.

6.1 Explanation of Ultrashort Laser Pulse

The very first laser system was fabricated by T. Maiman [26] in the year of 1960 with a power of 10^6 W/cm² and since then; enormous progress made possible the laser field intensity goes up to 10^{22} W/cm². An ultrashort optical pulse may be described as a few-cycle pulse [23] operating near the Fourier transform limit in which the temporal envelope exhibits rapid amplitude variations over a duration (Δt) that is much shorter than the intrinsic relaxation times (τ) for most of the materials. Few cycle laser is generated through a process called mode-locking [27] in which light pulse hosting large number of modes (Integer multiples of the half the wavelength (λ)) superpose coherently.

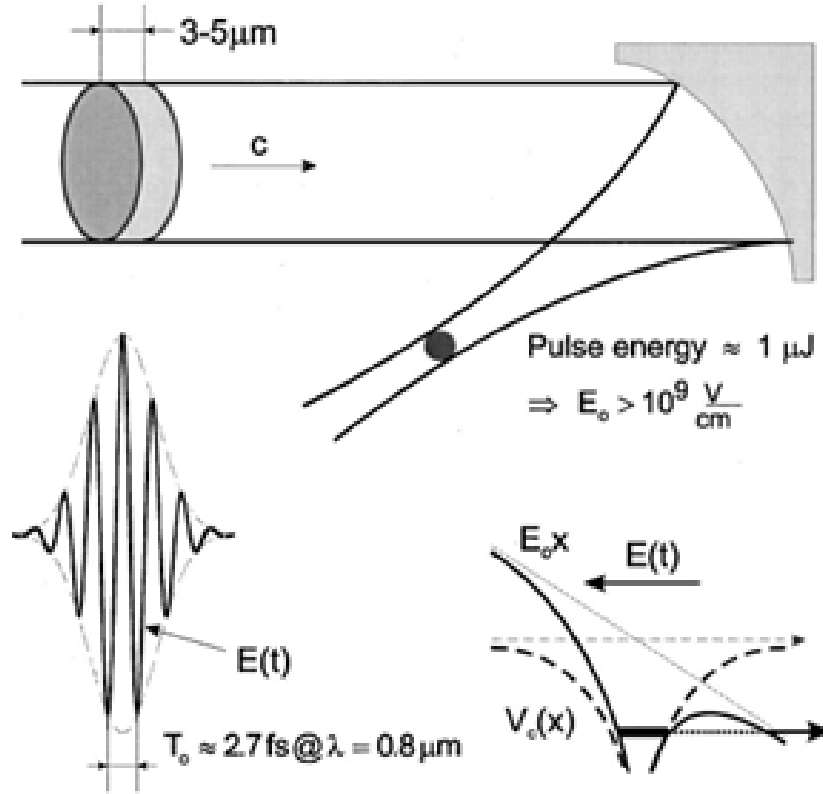


Figure 11: Intense few-cycle ultrashort pulses originates from high-degree of spatial and temporal coherence [18]

—————theory—————

6.2 Femtosecond pulse generation process

6.2.1 CW Pump Laser

Femtosecond pulse generation process begins with the CW green laser which is used to pump the gain media of the oscillator and the amplifier simultaneously. The Verdi V18/V20 interface is based on optically pumped semiconductor laser gives high-power continuous green laser at 532nm which is the core part of the few cycle pulse generation method provides stable energy output to function the Titanium-Sapphire (*Ti : Sa*) gain media [28]. It pumps the oscillator by supplying the CW power to the Ti:Sa crystal to maintain mode-locking together with pumping the amplifier (RegA), sustaining high-energy input needed for the Regenerative Amplifier to boost the seed pulses into high-energy pulses [30]. Fundamentally the functioning of the Verdi G-series is non-identical from the usual Diode-Pumped [29] Solid-State (DPSS) laser system, rather it uses a thin chip ($< 10\mu\text{m}$) based amplifying media

that helps to removal of heat from the chip that is essential to maintain beam shape from distortion. Since there is no thermal lensing, the Verdi can be tuned from 10% to 100% with power range 2 W to 20 W without any interruption in beam size or divergence changing. The following figure (14) provides a schematic representation of the Verdi system's physical layout.

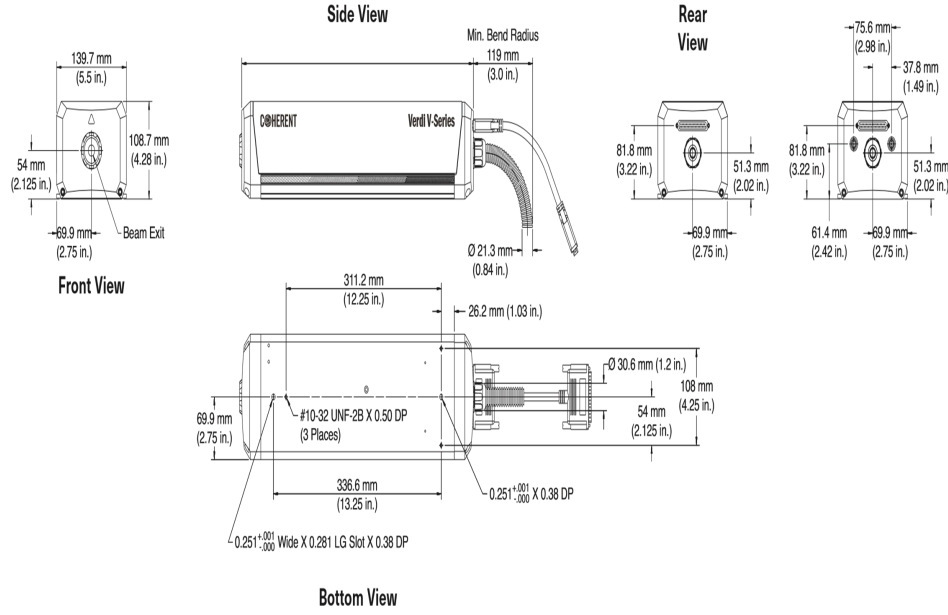


Figure 12: Architectural Overview of the Verdi V-Series Interface[[28](#)]

The operating wavelength of 532 nm (Power 13.6 W) from the Verdi V18/V20 is attained by means of cavity-enclosed frequency doubling (Second Harmonic Generation) process by using a near-infrared beam of 1064 nm. The output beam maintains a nearly TEM₀₀ Gaussian profile [\[28\]](#) with a beam quality factor, $M^2 < 1.1$ and this allows the beam to get focused into a tiny spots within the crystals which in returns maximize the efficiency of the pulse generation.

6.2.2 Titanium-Sapphire (Ti:Sa) Oscillator

The core part of the femtosecond pulse generation is Titanium Sapphire Oscillator introduced for the first time by Mouton [\[31\]](#) in the year 1982 at Lincoln laboratory [\[34\]](#) which opens window for generation of high-energy pulses based on mode-locked technic. By following this

process, different frequencies or color components (longitudinal modes) forced to oscillates by maintaining fixed phase relationship inside the resonator cavity. The CW green laser beam 532 nm is used as seed pulse for the oscillator's operation and pulse generation process involves various stages including regenerative amplifiers which amplify the strength of the pulse, then stretching and compressing the pulse with time and frequency domain followed by optical parametric [32] amplification (OPA) process. This oscillatory system can produce pulses as short as 30fs with wavelength centered at 800 nm with pulse power 400 mW and pulse emission rate 74MHz. The tuning ranges comparing with other materials is presented [34] graphically in figure (15).

Due to having extensive gain [34] bandwidth (660 to 1180 nm or 15200 to 8500 cm^{-1}), Titanium-Sapphire $\text{Ti:Al}_2\text{O}_3$ crystals are the most preferred gain media for generating tunable (partial spectral tunability is 57%) and ultrashort pulse lasers [33]. Although, the fact is that the popularity and uses of this crystal remains mainly confined to basic research because of possessing high loss; practical and industrial applications still requires to improve it's efficiency by using commercially available components. Gaining high efficiency not only helps to lower the required pump power but also facilitate reducing the system's size and cost with enhancing its reliability.

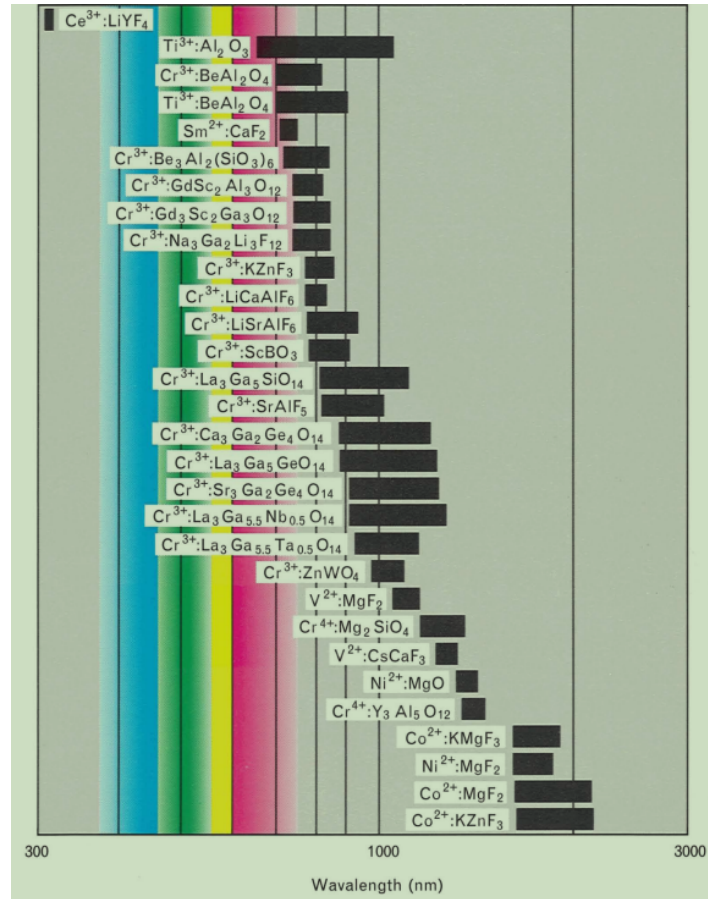


Figure 13: Graphical overview of variable solid state lasers [34] and corresponding operating ranges where it is clearly visible that Titanium-doped sapphire(Ti:Sa) has ultra-wide band-width spanning visible to near-infrared region of the spectrum than other material groups.

This lasers system can achieve pulse widths near the theoretical bandwidth limits (70%) and [33] conversion [33] efficiency (optical to optical) is factually very low; often with only a few percent owing to high crystal losses in addition to a low product of emission cross-section (σ_{em}) and fluorescence lifetime (τ). As a laser host material, $\text{Ti:Al}_2\text{O}_3$ provides as an exceptional crystal for lasers like the ruby laser because of possess superior physical and optical properties. It has Mohs hardness scale 9 out of 10 (Diamond) makes it ideal for maintaining high-quality, scratch-resistant surfaces with high thermal conductivity similar to copper at 80 K [34]. This remarkable aspects favor to gain high average powers without any thermal degradation. Subsequent figure (16) describes the working method of $\text{Ti:Al}_2\text{O}_3$ laser oscillator system where the crystal sits in the middle of the cavity.

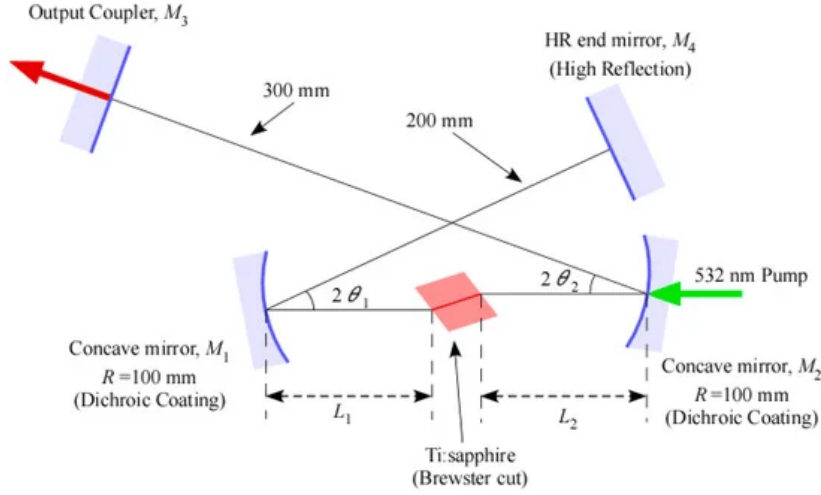


Figure 14: Titanium sapphire oscillator system which involves multiple mirrors and a gain medium made with Titanium-doped sapphire(Ti:Sa) crystal [33]

In this special optical setup, the CW pump beam get focused by using pair of mirrors. A Brewster cut titanium(Ti^{3+} doped sapphire crystal having a 0.25 wt.% concentration and 5mm length is used and the crystal itself placed on a radiator made of copper which helps to maintain standard room temperature ($20^{\circ}C$). The figure of merit (FOM) is in the order of 200 which quantify the performance factor of the laser medium. The oscillator system made of two concave dichroic coated mirrors M_1 and M_2 (Curvature $R = 100$ mm), an end mirror M_3 with higher reflectivity and a output coupler plane mirror M_4 . Since the mirror is coated with dichroic layers, it reflects certain wavelength of light and the initial pump beam can only passes once through the laser gain medium. For single pass pumping, the crystal's absorption efficiency rate (η_a) at the pump wavelength of 532 nm is measured approximately 90.5%. The efficiency (η_a) boils up to 99.1% when the pump beam is passed twice through the gain medium [33].

6.2.3 Principles of Mode-Locking

There are different techniques that are used to generate very ultra-short pulses and mode-locked mechanism is one of the best way of doing it. Gürs and Müller worked on the very first mode-locked ruby lasers laser system, Statz and Tang He-Ne lasers. The way how this mechanism works was written by DiDomenico and Hargrove et in 1964 .Yariv and Hargrove et al overcome the resonator loss for the first time through active mode-locking while using a saturable dye initiate the laser (Ruby laser) mode-locked introduced by Mocker

and Collins [47]. The very first practical application of self-mode locking procedure was in 1991 implemented while working with femtosecond pulse generation by Sibbett group which is famously known as kerr-lens mode locking [44]. In any standard laser system, such as continuous wave (CW), free-running or Q-switched laser, the output originates from the impinging of several discrete longitudinal modes or frequencies [43, 45]. Inside the resonator; various modes oscillate concurrently. The specific number and type of these modes are determined by the cavity's resonance frequencies and its length. Due to the random phase and amplitude of each frequency, the total intensity distribution inside the cavity appears as inherently random [42, 43]. Only the modes with the strongest gain exceed the oscillation threshold and the net gain is quasi-monochromatic light emission. Furthermore, ultrashort pulses in the picosecond to femtosecond range is generated by employing the mode-locking technique [42]. Mode-locking is a primary physical process that alters the output of a laser by utilizing a continuous monochromatic beam into a series of intense yet ultrashort light pulses [43].

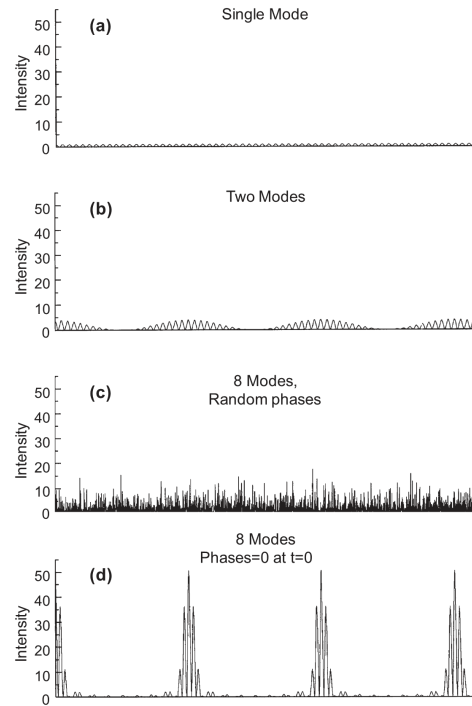


Figure 15: Overview of random oscillating longitudinal modes inside the the resonator cavity. The effect of multiple mode addition on phase [41]between the modes are shown in (a) -(c) and phase [41]between the modes and its consequences giving a series of ultimate pulses shown in sub-figure (d)

This process (Mode-Locked) adds an extra effect by establishing a fixed phase relationship (synchronizing pulses to the same phase factor or to a common value) between the longitudinal modes within the medium's fluorescence bandwidth. Consequently, this random fluctuations of the laser are pushed through collective coherence. At the end, this constructive interference transformed into a single, high power pulses with durations picoseconds down to femtoseconds [42, 47] range. The transformation of this revolutionary methods in laser physics since 1964 has established two primary methodologies. One is active mode-locking in which the synchronization is done by using an external device which is called modulator. On the contrary, passive mode-locking that relies on the utilization of internal nonlinear elements like saturable absorbers to automatically shape and shorten the pulse [42].

over the decades, these mode-locked technique have been modified to diminished the physical limitations like gain bandwidth and dispersion. Ultimately this favors the generation of ultra-fast light that can probe nonlinear responses in various media [47]. These days, mode-locked lasers are essential tools across the every aspects of human life and serve as the technical tools for novel inventions like three Nobel Prize winning breakthroughs: optical frequency combs, chirped pulse amplification and atto-second light pulses [42]. Beyond of this, its ability to deliver strong peak intensities with high temporal precision has made it indispensable in applications including spectroscopy, metrology and medical procedures such as refractive surgery. [42, 47]. The following figure (18) exemplify the time evolution of laser in different stages for different materials over decades.

6.3 Active and Passive Mode Locking

6.3.1 Chirped Pulse Amplification (CPA)

The acronym CPA stands for Chirped Pulse Amplification and this technique was developed in 1960 to enhance the power of radar at Rochester University by Donna Strickland and Gérard Mourou in the year 1980 [35]. The radar operation back then hindered due to unavailability of high energy ultra-short pulses and having high pulse intensities GW/cm^2 without any circuit induced serious destruction of the amplifying medium via self-focusing while pulses amplification. The power of the ultrashort laser pulses generated from the oscillator system is very low ($\sim 10^{-9}$ J) with a timescale on the order of 10^{-12} to 10^{-14} s and repeating rate 10^8 s^{-1} [31]. There are distinct methods [37] are being used to amplify the optical pulses

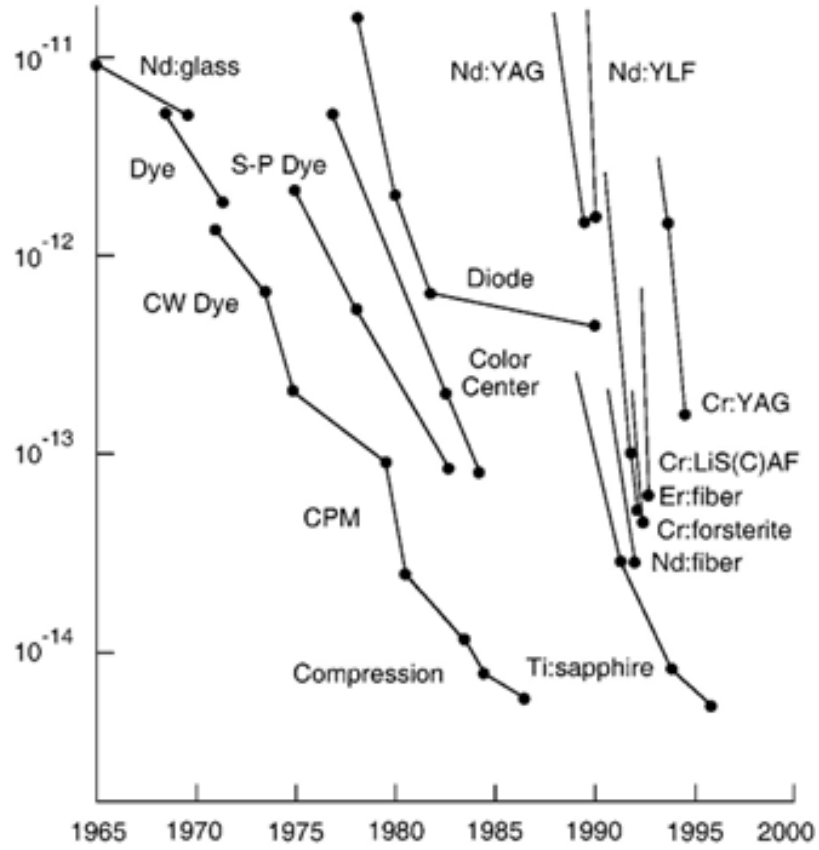


Figure 16: Representation of laser advancement dated back its invention year and Pulsewidths of various laser systems [44, 47]

by keeping the pulses very short at the same time [31]. These process are Chirped pulse amplification (CPA), Optical Parametric Amplification (OPA) and Optical Parametric Chirped Pulse Amplification (OPCPA) [35, 37] used to increase to pulse energy up to *mJ* level. The amplification process based on to stretching out the pulses on time and frequency domain prior to amplification succeeded by compression [35]. To raise the peak power of the ultrashort pulses, several conditions must needs to satisfy by the gain medium. The amplifying capacity of the host crystal or medium needs to sufficient enough to amplify the spectrum at broader ranges and removed the stored energy from the gain medium [36, 39] properly. The pulse energy always needs to be within the limit so that the peak intensity does not cause the B-integral to exceed critical thresholds to prevent beam degradation and optical damage.

$$B = \frac{2\pi}{\lambda} \int \frac{\Delta n}{n} dl = \frac{2\pi}{\lambda} n_2 \int_0^L I(z) dz. \quad (26)$$

cite M45 The following graphical representation of CPA showed in figure—technic where a beam of short low energy pulse get spread out with the help of dispersive medium before

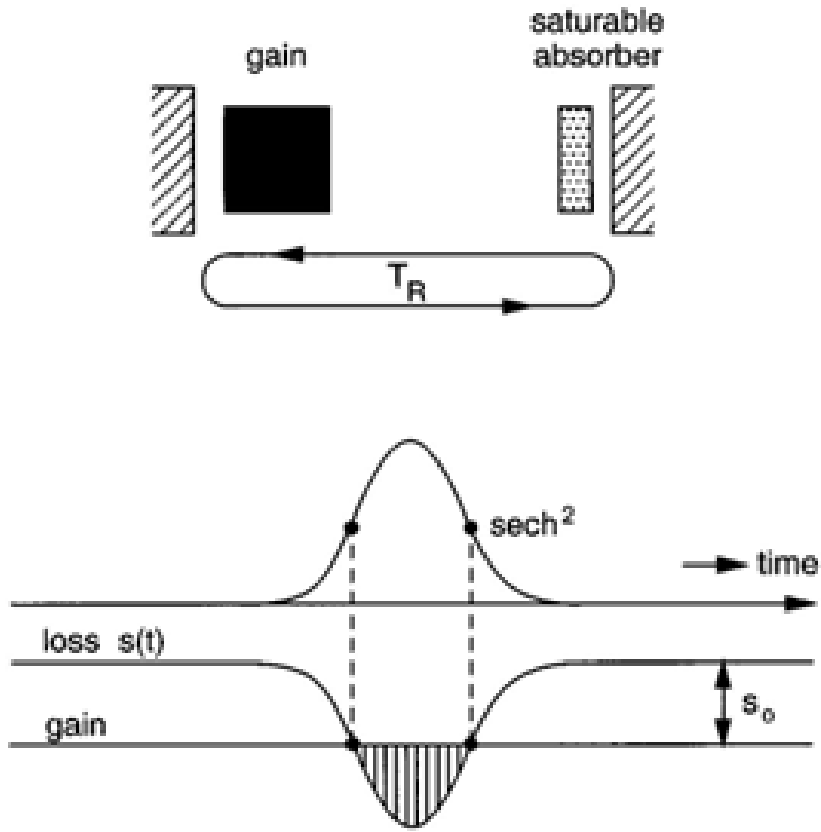


Figure 17: []

putting the pulse into the amplifying medium on time domain [35]. The special design of the dispersive medium or grating pairs directs the high frequency light to follow the longer path than the low frequency part. This adds a positively chirped effect where high frequency lags behind than low frequency part with longer propagation time by an order of magnitude $10^3 - 10^5$ relative to the initial pulse.

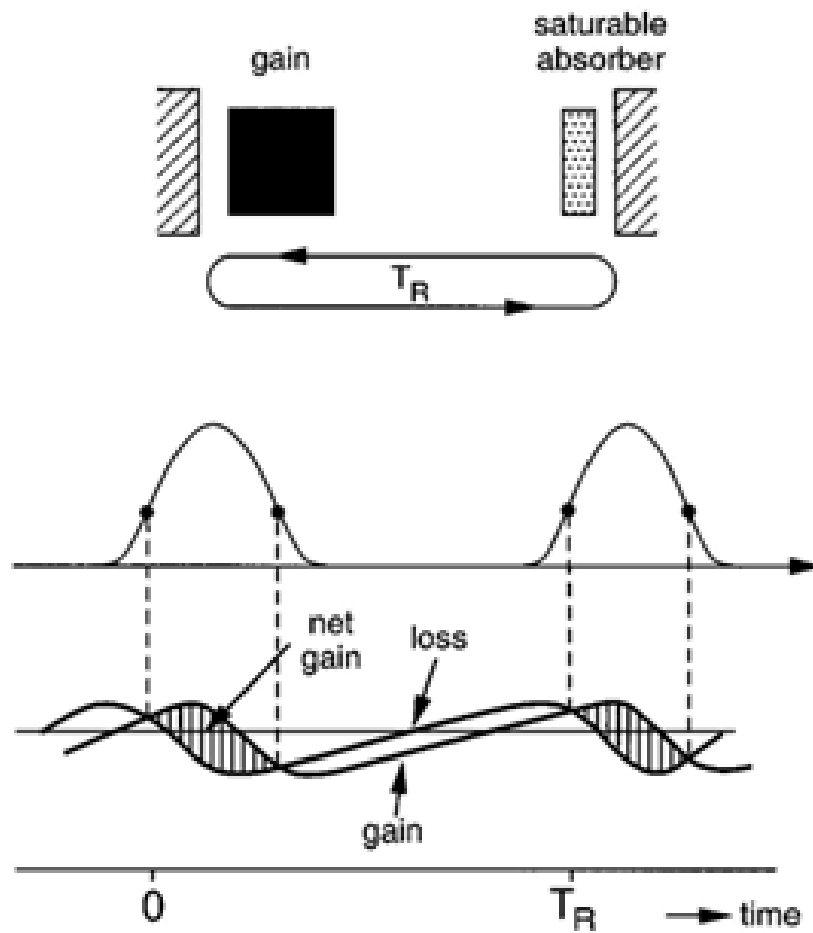


Figure 18: []

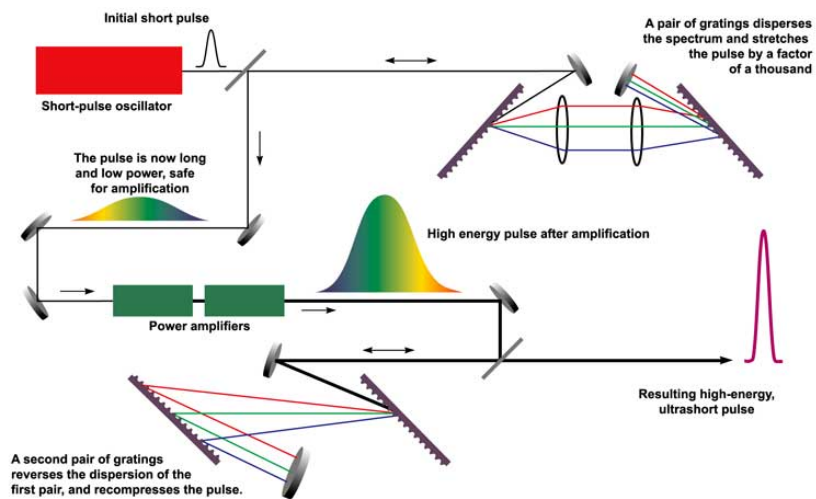


Figure 19: Overview of the amplification process and Sequential stages through chirped pulse amplification for having high energy ultrashort pulses [31,35]

Stretcher design

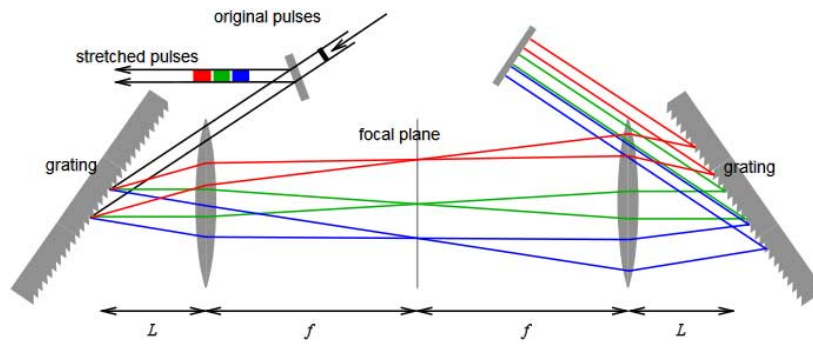


Figure 20: [33,35]

RegaA and multipass amplification

compressor design

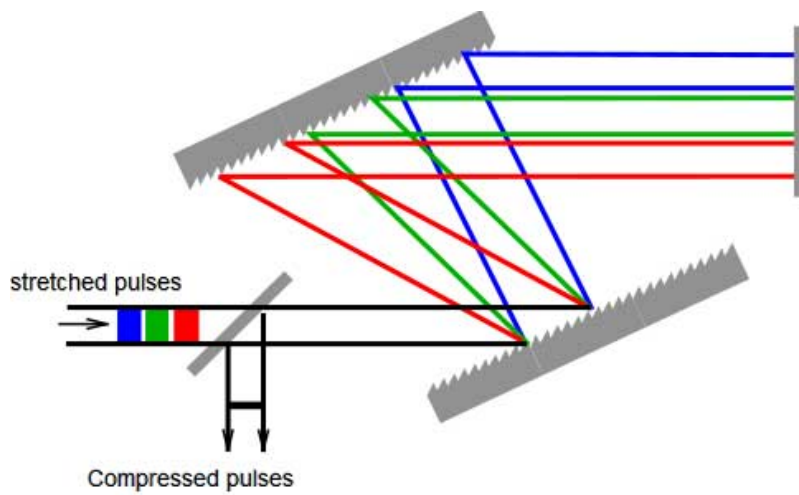


Figure 21: [33,35]

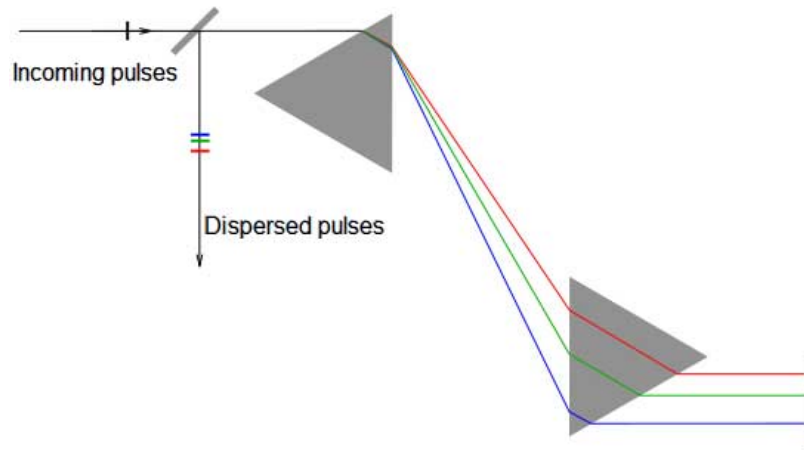


Figure 22: [33,35]

6.3.2 Optical Parametric Amplifier

7 Super-Continuum Generation (SCG)

Super-continuum Generation (SCG) is a special nonlinear optical process in which a intense narrow-band laser pulses undergo into an extremely broad continuous spectrum of light spanning multiple octaves [2]. The spectral ranges extending from the ultraviolet (UV) through the visible (white light) and into the infrared (IR) regions by maintaining the high brightness, a high level of spatial coherence (allows tight focusing) characteristics similar of a laser beam, high output power, a broad and flat spectrum [21]. Coherent light source having a a single frequency is more than enough for various application [20] yet any light source that accompanies features of laser possess broader bandwidth along with extreme short pulse duration unveil era of prospect in variety fields of science, engineering, material science and spectroscopy. Specially, the supercontinuum light source drastically enhances the signal to-noise ratio (SNR) by lessening the acquisition time together with broadening the spectral limit [21]. In 197, Robert Alfano and Stanley Shapiro made one of the most remarkable discoveries regarding new type optical effects that give rise to frequency broadening through non-linear effect which is termed as Super-Continuum (SC).

8 DRS Experimental Setup and Components

The extended DRS (Differential Reflectance Spectroscopy) setup is a specialized optical technique aimed for high-precision monitoring of thin-film (e.g., perfluoropentacene (PFP)) deposition on crystal substrate in the forms of measuring its reflectivity. By analyzing the contrast between these two signal paths, this setup effectively determines the optical anisotropy and precise molecular alignment of the deposited organic material [3]. Rather than relying on a single beam measurement procedure, the system utilizes a dual-beam detection method to simultaneously capture reflected light intensity alongside the reference. This variable approach is particularly designed to resolve minute changes in the surface's optical response providing a high resolution real-time film history. The complete optical setup of the two-beam **DRS** measurement and standard **DRS** measurement with white light continuum (e.g., driven by femtosecond 800 nm pulse, discharge lamp) given in figure (23) and (24) depicts the key optical components and geometry. Through precise handling of the implemented optics, the contrast between these two signal paths effectively maximizes the efficiency of characterization process that might standard reflectance measurement procedures overlook.

The Pol-DRS [3] experimental setup includes several densely packed opto-mechanical layout to maintain interconnected beam paths specifically; the broadband light source (**non-polarized, UV-Vis-IR**), Spherical and Planar Mirrors, Coated Mirror (e.g., dielectric mirror, Al mirror, Au mirror), Iris, Filter, Concave and Convex Lens, Beam Splitter (non polarizing cube or basic beam splitter mirror), Sapphire Crystal, triggering and the detection systems (function generator, spectrometers, data processing software).

Since the experimental setup is designed for high resolution reflectance measurements, so a stable and high intensity broadband light source generally from a LASER (additionally with discharge lamp) laser driven super-continuum light source that begets through the interaction with sapphire crystal is used to provide light across the desired spectral range (wavelength ranges from **200 nm - 1000 nm**). The reason of choosing broad white light continuum is that it gives much freedom of having insights of reflectivity or investigating surface properties at wider ranges of wavelength at once rather examining at a single wavelength separately. The deployment of all the associated optics together with the manipulation and steering of light source is described in the following paragraph.

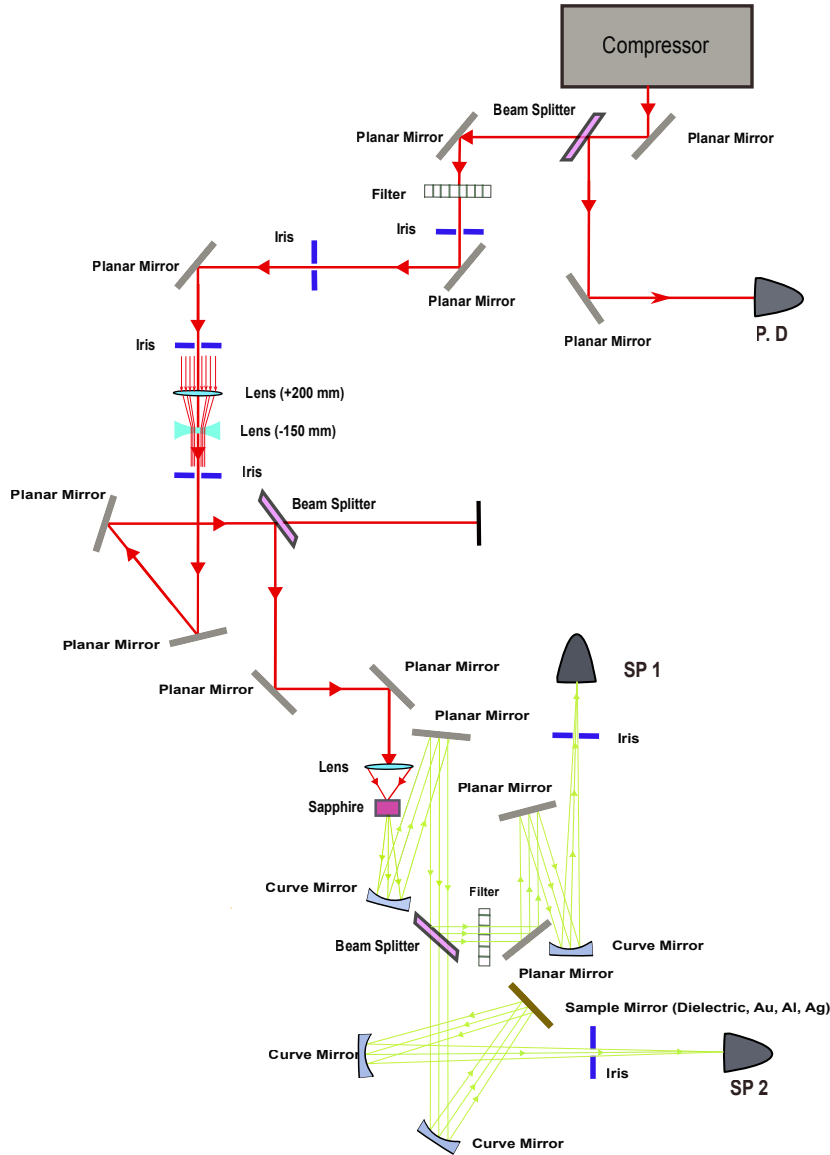


Figure 23: Schematic illustration of two beam differential reflectance measurement experimental setup. The main beam path accommodate sample while secondary beam path represents reference path.

The experimental procedure begins through the generation of femtosecond pulses centered at **800 nm** originating from a Ti:Sapphire regenerative amplifier and ends at compressor compartment to boost its peak intensity (observed power from the compressor is **116 mW**). Setup and used opto-mechanics can be alignable to any orientation depending on user interest and for specific experimental goal. The femtosecond pulses is redirected at 90° by placing a planar mirror in laser's beam path by maintaining at an angle of 45° respect to the incoming beam right after the pulses leave the compressor section. This redirection of the beam just

simplifies the design and allows for versatile use in relatively shorter spaces. A primary beam splitter splits((50:50) the initial beam into two secondary pulses, guiding a portion of the pulses by using a planar mirror to a Photodiode (P.D) for pulse and power monitoring. On the other hand, the main experimental path for broadband light generation; a series of planar mirrors, irises and filter is deployed to ensure precise alignment and beam control before passing through the telescope-system. Planar mirrors are universally used optics for laser beam re-direction because it offers precise control while minimizing distortions, power loss and high damage threshold (planar mirrors are coated with dielectric substances which gives reflectivity reaches to almost 100%). Since its surface is flat; it works based on the law of reflection , so the parallel beam remains parallel (collimated) after reflection. To control the beam's intensity without distorting the spectral property, an adjustable neutral density filter wheels is used that prevent damaging of sensitive instruments (e.g., spectrometers). For controlling beam's size and diameter, iris is primarily used in a femtosecond laser path. By closing or enlarging the iris down to a small or larger aperture , it can ensure the beam passes through the exact center of lenses and crystals. Iris is also used to maintain a standard gaussian beam profile by clipping the outer edges of the beam and by removing unwanted stray light.

It very common in femtosecond laser operation to use a pair of lenses (**telescope-system**) for controlling the spot size and divergence. The relation between the beam's diameter and spot size is inversely proportional [16,41] which can be written as:

$$d = \frac{4\lambda f M^2}{\pi D} \quad (27)$$

Where,

- d = focal spot diameter
- λ = wavelength of the laser light
- f = focal length of the focusing lens
- D = diameter of the laser beam at the lens
- M^2 = beam quality factor

So increasing beam diameter reduce the spot size increasing the beam divergence along its propagation. There are two types of telescope system that can be used to control the spot size

which acts either as a beam expander or reducer. **Keplerian telescope system** is better for increasing the beam convergence (the beam remains collimated over long distance) which consists of a long focal (convex lens) objective lens ends with short focus (convex lens) eye-piece lens. Since the experimental goal doesn't demand any intermediate focus, so it's not suitable for the experimental setup, rather **Galilean telescope** system is most effective which avoids intermediate focus along with keeping less beam divergence (since the used optics are connected closely). So lenses combination consisting of a objective lens (**convex lens, +200 mm**) followed by an eye-piece lens (**concave lens, -150 mm**) collimates and reduce the beam diameter to **0.75%** of its initial diameter. As soon the femtosecond laser beam leaves the telescope system, the beam again pass through another beam splitter (the reason for using beam splitter is at this point is to multipurpose use of the femtosecond pulses) after it gets re-directed through a pair of planar mirror. The primary beam path is maintained towards the super-continuum generation while the other portion (can be used to pulse monitoring or **Harmonic Generation (HG)**) is blocked by a beam blocker. Another two planar mirrors has been used for beam re-direction by following the above mentioned condition and pass through a short-focal length (focusing mirror, **f=50 mm**) convex length. As the objective of the experiment is to measuring reflectivity using super-continuum, so a non-linear crystal (Sapphire) is placed at the focal point of the converging lens. The **800 nm** femtosecond beam is focused tightly onto a Sapphire crystal. Due to high intensity, non-linear process happens (discussed in chapter 3.3 and 7) resulted in broad-band light.

The produced super-continuum light get diverge soon after leaving the crystal, so the beam has been re-collimated using a concave mirror (high reflective Aluminum mirror) which then redirected using a planar mirror. The standard DRS comprises of two beam configuration and owing to this, a non-polarizing cube beam splitter (**(50:50)**) is used again which creates a reference beam and a beam for the sample path. Same number of planar and curved Aluminum-coated (high reflectivity across broad spectral range) mirror is used both in the reference and sample beam path to avoid any unwanted effects.

The reflected beam from the beam splitter is passed through a filter to control the intensity and re-directed by using two-planar mirrors followed by focusing the beam onto spherical mirror which is called focusing mirror. The spectrometer is placed at the focal point of the

focusing mirror so that the collimated beam directly hits the entrance slits of the spectrometer (**SP 1**). While the transmission beam from the beam splitter redirected and focused onto the desired sample (e.g., Dielectric, Aluminum, Gold, Silver) mirror after it reflects from the curbed mirror. The purpose for using a curved mirror at this point is to control the dispersion and proper focusing the continuum source onto the sample mirror . The incoming beam from the sample mirror gets reflect from the sample mirror and remains almost parallel after reflection. Then the reflected beam is rand tightly focused to the entrance slit of the sample spectrometer (**SP 2**)

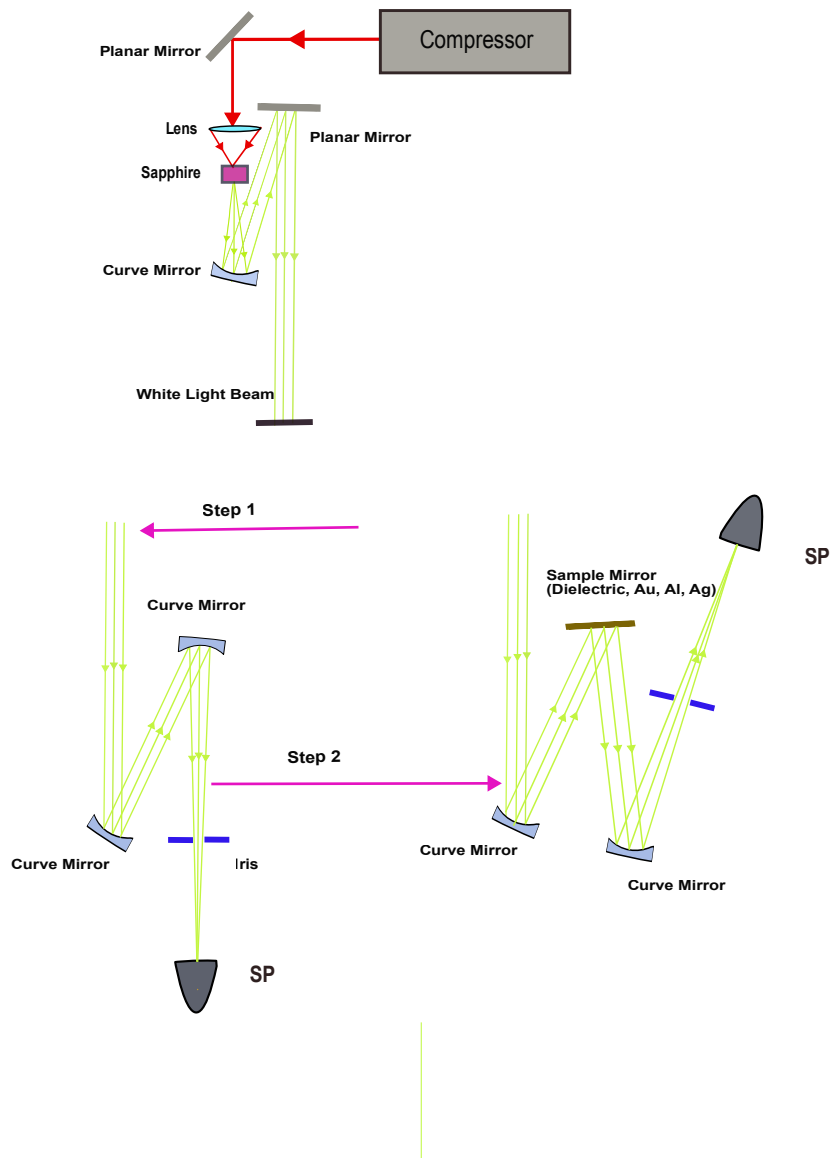


Figure 24: Schematic illustration of standard reflectance measurement experimental setup

$$\frac{\Delta R}{R} = \frac{R_{\text{sample}} - R_{\text{dark}}}{R_{\text{reference}} - R_{\text{dark}}} \quad (28)$$

Two highly sensitive Avantes (Serial:1907283U1 and 7513481SP) has been used to measure the intensity of the white light. Since the experimental goal is reflectance measurement, one spectrometer used for collecting reference spectrum and the other for reflected light from the sample spectrometer. Avasoft 8 application is used for real-time spectra acquisition, background and dark signal is subtracted by the application during the measurements. Most of the data acquisition is done by setting the integration time 20 ms and for a scan rate 2 (can be adjusted depending on the light properties). During the reflectivity measurements of the dielectric mirror (lab 3), the integration time is set 4 ms for reference spectrum and 12 ms for sample spectrum just to have a smoother spectrum (variation of the integration time solely depends on photon flux of the light source).

In a high-speed measurement system, light sources fluctuate due to thermal drift or signal modulated due to electrical noise of the spectrometers. These fluctuations or differences appear in the measured data if the two channels aren't perfectly synchronized. The level of sensitivity of the detectors system is different which also broadly affects the spectrum. These fluctuations or differences appear in the measured data if the two channels aren't perfectly synchronized, so a function generator (not included in the experimental setup) is used during the measurements for triggered acquisition. Triggered acquisition is vital in multi-beam system measurement because it forces the channels (sample and reference) to record data at the exact sametime which cancel out all the environmental and system instability ensuring $\frac{\Delta R}{R}$ perfectly represents material's optical properties.

With the help of high-speed electronics systems including spectrometers and PC, Avasoft 8 application records the intensity data of sample ($I_s(t)$) and reference ($I_r(t)$) as a function of wavelength (nm). Lastly the Reflectivity measurement is done by using the formula given in equation (28). A possible experimental setup for standard reflectance measurement procedure is shown in figure (24). The operation is as same as before with an exception of one beam path rather than having two beam paths at the same time. In this setup, the intensity of light is measured using only two curved mirrors followed by using the sample mirror and

taking the intensity measurement again. From these two measurements, light intensity can be subtracted leads to the intensity only for sample mirror. It is important for comparing which setup performs well which ultimately indicates mirror's performance.

9 Reflectance Measurement and Data Analysis

9.1 Reflectance Measurement

Differential Reflectance Spectroscopy (Pol-DRS [3]) is an advanced technique and a precise measurement procedure comprises a specialized optical setup designed to measure the changes in the intensity of reflected light resulting from the presence of any sample or molecular layer. Factually, it quantifies how much a surface's optical properties changes relative to its reference state [4]. The measurement begins with measuring reflected beam intensity of the clean, unmodified sample surface (the substrate or thin-film) across the desired wavelength ranges before and after modification or simultaneously using two beam- configuration. The sample can be located at any direction by adjusting optical setup, the reference and reflected signal then collected by two spectrometers. By comparing the ($\Delta R/R$) ratio signal over time directly linked to the thickness of the deposited film or film's properties where the peaks indicate characteristic electronic transitions.

9.2 Data Analysis

9.2.1 Lab 1 (light Source: fs LASER driven continuum and dielectric mirror)

The objective of obtaining a sample (spectrometer 1907283U1) spectrum alongside a reference (spectrometer 7513481SP) spectrum is usually to perform a referencing operation to correct the sample data for that light source properties and background signal. The spectra; sample (blue) and reference (red) represented in figure (25) were obtained by deploying two Avantes spectrometers and a broadband white light source. In terms of background noise, both the reference and sample spectrum exhibits low baseline (Presented in figure (26)) typically below 600 AU ; sample scaling around 394.1 AU and reference scaling around 541.6 AU across the entire wavelength range. Tracing and correcting baseline or the signal floor is extremely important in spectroscopic data analysis because it facilitates by giving qualitative along with quantitative idea about the data by setting the intensity values relative to a zero

point.

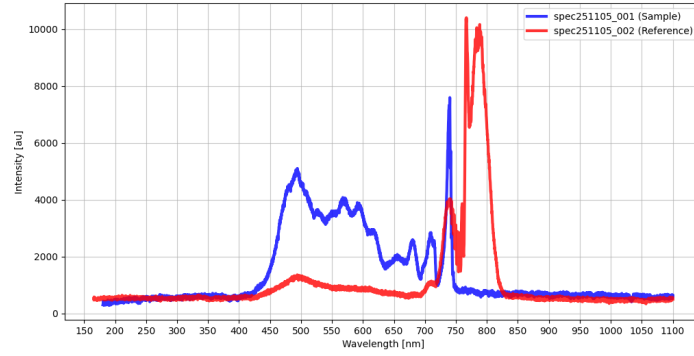


Figure 25: Relative Intensity mapping of Sample (blue) and Reference (red)

Concerning the reference spectrum, the background offset is exactly around ~ 394.1 AU and looks almost flat across the entire spectral range which a bit fluctuation around ~ 350 AU and this could be due to external influence which distort the spectrum at this point. Aside the flat baseline infers that the instrument is stable plus well calibrated and the data is highly reliable. On the other hand, the background offset for the sample is around ~ 546.6 AU which is ~ 147 AU difference between the baseline of sample and reference. Since the absolute intensities of the peaks are really essential for uncovering reflectance ratio, subtracting the baseline offset from every data point from the reference and sample spectrum sets the common baseline offset at 0 AU which can be seen in sub-figure (26). The (offset) flooring formula for baseline correction is as follows:

$$\text{Corrected Signal}_{(\text{sample, reference})} = \text{Sample Measurement} - \text{Dark Signal}$$

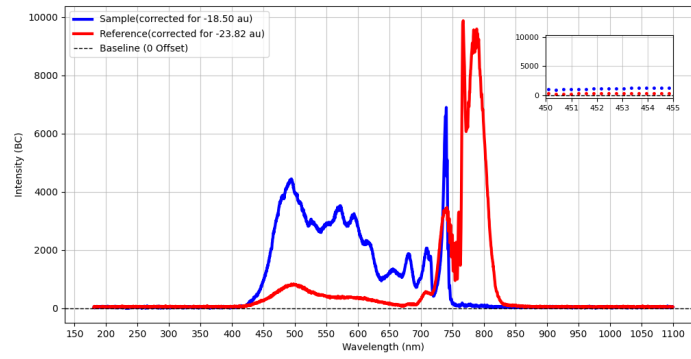


Figure 26: Combined background corrected spectra both for sample and reference

The maximum intensity counts for the sample channel is 299.50 (au) with 318.00 (au) for dark counts which in turns a baseline correction exactly -18.50 (au). Contrary, the maximum dark counts for the reference channel is 524.82 (au) and intensity counts is 501.00 (au) gives a baseline inset -23.82 (au) which is corrected and can be seen from figure (26). The reference spectrum presented in the figure (25) is obtained through using white light continuum driven by femtosecond laser pulses. It is certain that there are broad intensity distribution can be observed across the spectrum (red) with wavelength centered around 800 nm. On the contrary, the sample (blue) spectrum (dielectric mirror was used to obtain the sample spectrum) showed a broad complex peaks in the visible region centering between 450 nm and 750 nm indicates that the sample is reflecting significantly more light in this blue to orange wavelength range compared in the UV and NIR ends of the spectrum (below 450 nm and above 750 nm). The highest reflection occurred around 750 nm which is almost twice in magnitude compared to photon intensity in the visible range. In addition, the sharp peak (reflectance) intensity observed between 556 nm to 600 nm (Vis) and at 680 nm is likely due to the sample material interacting with the light source strongly and the reflection of photon at these wavelength is maximum compared to the NIR and red part of the spectrum

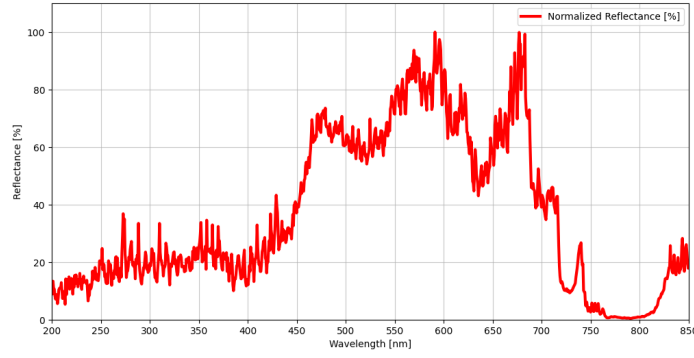


Figure 27: Normalized Reflectance Ratio Relative to Reference spectrum (Continuum)

The plot for reflectance ratio are presented in figure (27) by setting the sample's data in the limit of (0,3184) with a wavelength range is from 180.53 nm to 1100.06 nm and reference data (71,3255) with a wavelength range from 165.69 nm to 1100.20 nm. Another reflectance ratio plot is also shown in figure (28) in the wavelength range 400 nm – 800 nm.

For finding **reflectance ratio**, the very first step is to calculate the raw ratio (how strong the sample's intensity relative to reference intensity) relative the sample's signal count to the reference's signal counts between adjacent wavelength. For simplicity, wavelength 440nm has been taken for example where the reference and sample spectrum doesn't share any common wavelength that this data point (since spectrometers don't collect data on a continuous and infinite manner rather at discrete intervals). The linear interpolation formula used for commonize data points is given in equation (29).

$$y = y_0 + (x - x_0) \frac{y_1 - y_0}{x_1 - x_0} \quad (29)$$

Where,

- x = target wavelength (nm)
- y = intensity at target wavelength (counts)
- x_0 and x_1 = nearest measured wavelengths (where $x_0 < x < x_1$)
- y_0 and y_1 = measured intensities at those respective wavelengths.

Interpolation creates a virtual data point exactly at 440.00 nm for both channels resulting in a common wavelength grid. The intensity counts for the sample spectrum (adjacent to 440 nm) is 1356.50 au at 439.81 nm and 1335.50 au at 440.11 nm respectively with interpolated intensity 1348.50 au. On the other hand, the intensity counts for reference spectrum (adjacent to 440 nm) is 652 au at 439.72 nm and 644.50 au at 440.02 nm with interpolated intensity 651.97 au. Now the raw ratio can be obtained by relating the sample's signal count to the reference's signal count at a certain wavelength which is shown following.

$$\text{Ratio}_{(440) \text{ nm}} = \frac{1348.80}{651.97}$$

So the raw ratio is approximately "

$$\text{Ratio} \approx \mathbf{2.0776}$$

After having the raw ratio, the next step is to find normalization factor which removes experimental artifacts like light fluctuations and sample concentrations to certain points by scaling data relative to maximum data point. The maximum raw ratio which is aboserved 7.0783 at 680 nm along with sample and reference count is 1827.26 and 258.15). The normalized ratio can be obtained by dividing the raw ratio by the maximum ration value for that specific wavelength,

$$\text{Rationorm}(440 \text{ nm}) = \frac{2.0776_{(\text{ratio at } 440 \text{ nm})}}{7.0783_{(\text{ratio at } 680 \text{ nm})}}$$

$$\text{Ratio}, \mathbf{R} \approx \mathbf{0.0.2935}$$

Lastly, presenting this value in percentage gives the desired reflectance ratio in 100% scale

$$\mathbf{R}_{\text{ratio}} = 0.2935 \times 100$$

$$\mathbf{R}_{\text{ratio}} \approx \mathbf{29.35\%}$$

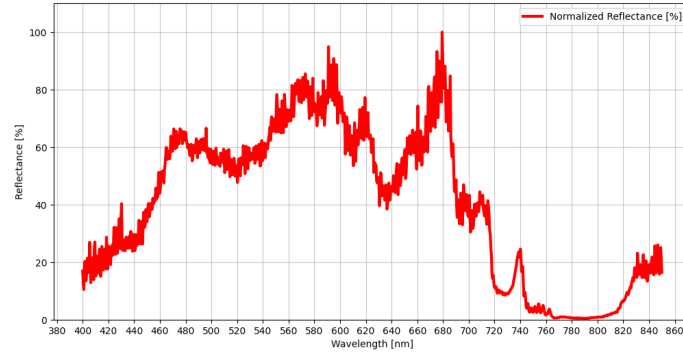


Figure 28: Reflectance Ratio of Dielectric Mirror, Wavelength Ranges [400nm, 800 nm]

The reflectance ratio plot (figure-28) illustrates the reflectances spectrum of a dielectric mirror and quantifies how efficiently the mirror reflects light across different wavelengths ranges. The most prominent features can be noticed in the plateau of the middle of the graph (roughly between 400 nm and 700 nm). Highest reflectivity is observed in yellow-orange region having a reflectivity around 98% and 95% with peak reflectivity 100% in the deep-red region. The reflectivity is significantly lower in the green and blue part of the spectrum which is a factor of two less compared to the yellow- orange region.

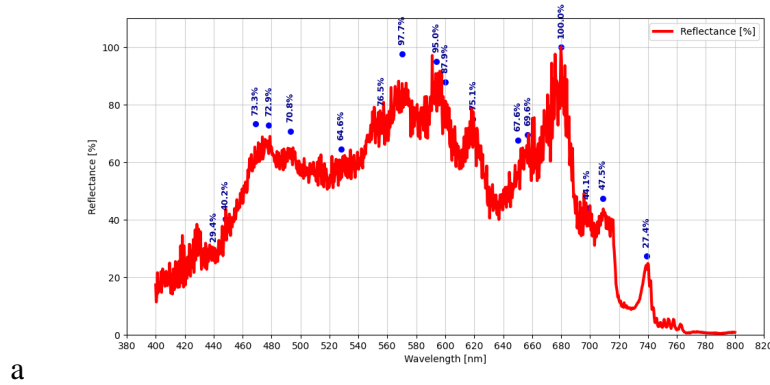


Figure 29: Reflectance Ratio With Peaks [Sliced, 400 nm - 800 nm]

The standard reflectance ratio measurement for two-beam configuration is performed via,

$$R_{ratio} = \frac{R_{sample}}{R_{reference}} \times 100\% \quad (30)$$

For having reflectance ratio of the entire spectrum range, the above mentioned formula and equation (30) is used and tabulated the reflectance ratio data (right side) in table (3).

Table 3: Dielectric Mirror's Reflectance Ratios at Different Wavelengths

Data submitted before				Error Corrected Data			
Wavelength. [nm]	Raw Ratio	Norm Ratio	Ratio [%]	Wavelength [nm]	Raw Ratio	Norm Ratio	Ratio [%]
440	1.5312	0.3291	32.91	440	2.0776	0.2935	29.35
469	3.6067	0.7752	77.52	469	5.1861	0.7327	73.27
478	3.9305	0.8448	84.48	478	5.1587	0.7288	72.88
493	3.8950	0.8372	83.72	493	5.0094	0.7077	70.77
528	3.3976	0.7303	73.03	528	4.5699	0.6456	64.56
556	4.1855	0.8996	89.96	556	5.4183	0.7655	76.55
570	4.4247	0.9510	95.10	570	6.9145	0.9768	97.68
594	4.4278	0.9517	95.17	594	6.7229	0.9498	94.98
600	4.1527	0.8926	89.26	600	6.2230	0.8792	87.92
619	3.6819	0.7914	79.14	619	5.3138	0.7507	75.07
650	2.7728	0.5960	59.60	650	4.7860	0.6761	67.61
657	2.9729	0.6390	63.90	657	4.9289	0.6963	69.63
680	4.0547	0.8715	87.15	680	7.0783	1.0000	100.00
698	2.3247	0.4997	49.97	698	3.1184	0.4406	44.06
709	2.6763	0.5753	57.52	709	3.3595	0.4746	47.46
739	1.8387	0.3952	39.52	739	1.9391	0.2740	27.40

The reflectance plot presented in figure (29) depicts the peak intensity value at different wavelength in the wavelength range 400 nm to 800 nm along with previously calculated data (table-3, left side) and reflectivity plot illustrated in figure (30). The previously calculated plot based on raw ratio counts at yellow-orange region while the current calculation is based on deep red part of the entire spectrum and this discrepancy originates from the interpolation just due to the

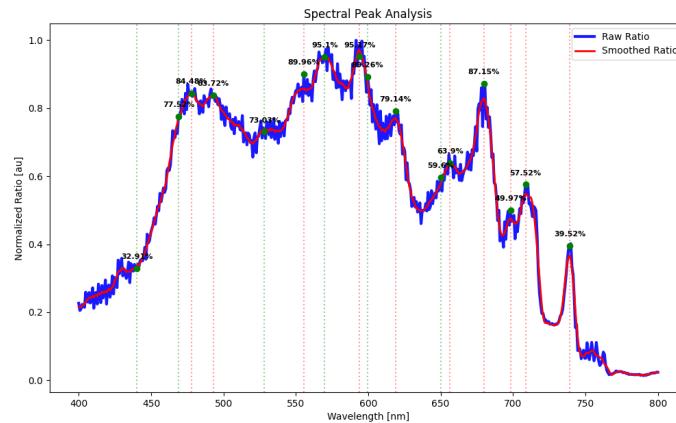


Figure 30: Spectral Peak Analysis in the Wavelength Range, 400 nm - 800 nm

9.2.2 Lab 2 (23.02.2026, Light Source: fs LASER, Sample: Al, Au)

Aluminum Coated Mirror

The consecutive second experiment has been performed by following the same principle described in lab 1. Femtosecond laser driven white light (continuum) source is used and sample mirror was replaced by Aluminum and Gold mirror. For relative intensity distribution of Aluminum (Al) mirror presented figure (32), it is obvious that the reference (white light) spectrum has almost symmetrical intensity distribution across the entire wavelength range (end of UV - end of NIR) with peak intensity centered around 620 nm (over 50,000 au). The sample spectrum showed also a symmetrical intensity distribution like the reference spectrum with having a peak intensity at 600 nm (over 50,000 au). There is a sudden decrease in the sample's intensity observed from 600 nm and a sudden deep drop observed at 623 nm with an increase upto 640 nm. Afterwards, a steady decrease is prominent upto 695 nm followed by sudden drop upto 710 nm. This fluctuation in the intensity is due to the absorption of light at this specific wavelength and which is most significant at 623 nm (deep absorption).

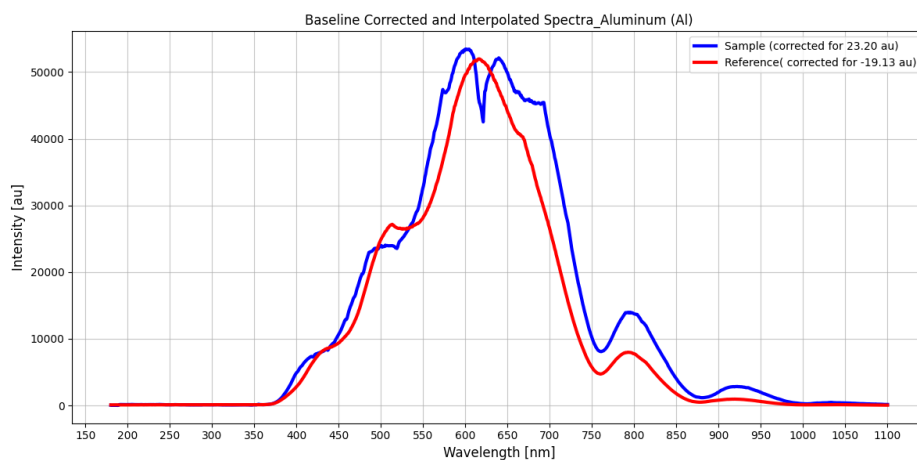


Figure 31: Relative Intensity distribution of sample (Aluminum) and reference

By closely inspecting the reflectance ratio plot (figure-32), it can be concluded that the Aluminum (Al) mirror has high-reflectivity across visible part of the spectrum. The reflectivity is most prominent at certain wavelength such as, 470 nm and 570 nm. The deep drop is also clearly visible at 620 nm and there is almost steady increase in the reflectance from visible to IR. The reflectivity in the IR or NIR part is more than double compared to visible part and reaches 98% at 980 nm. The subsequent figure (33) represents ratio relative to the ratio

at 790 nm rather than maximum ratio. There are less significant changes observed while normalizing respect to 790 nm. The ratio of reflectivity is almost same with a bit of increase for every wavelength. The ratio of the Aluminum reflectivity is tabulated both for maximum normalization factor and relative to 790 nm in table (4). The left side represents ratio respect to maximum ratio and right side is relative to 790 nm ratio factor.

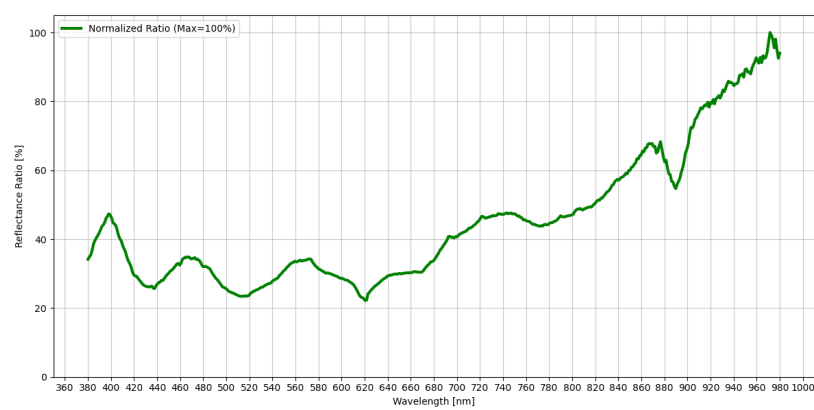


Figure 32: Reflectance ratio of Aluminum accross different wavelengths

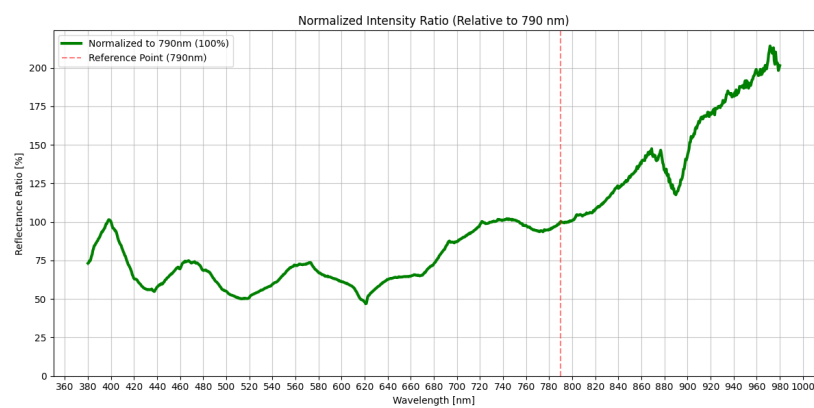


Figure 33: Reflectance ratio of Aluminum across different wavelengths relative to 790 nm

Table 4: Reflectance Ratios of Aluminum (Al)

Relative to Maximum Ratio				Relative to 790 nm			
Wave-length [nm]	Raw Ratio	Norm. Ratio	Ratio [%]	Wave-length [nm]	Raw Ratio	Norm Ratio	Ratio [%]
380.0	1.2852	0.3414	34.14	380	1.2852	0.7314	73.14
405.0	1.6189	0.4301	43.01	405	1.6189	0.9212	92.12
430.0	0.9933	0.2639	26.39	430	0.9933	0.5652	56.52
455.0	1.2001	0.3188	31.88	455	1.2001	0.6829	68.29
480.0	1.2050	0.3201	32.01	480	1.2050	0.6857	68.57
505.0	0.9209	0.2447	24.47	505	0.9209	0.5240	52.40
530.0	0.9803	0.2604	26.04	530	0.9803	0.5578	55.78
555.0	1.2303	0.3269	32.69	555	1.2303	0.7001	70.01
580.0	1.1788	0.3132	31.32	580	1.1788	0.6708	67.08
605.0	1.0548	0.2802	28.02	605	1.0548	0.6002	60.02
630.0	1.0011	0.2660	26.60	630	1.0011	0.5697	56.97
655.0	1.1340	0.3013	30.13	655	1.1340	0.6453	64.38
680.0	1.2728	0.3381	33.81	680	1.2728	0.7243	72.43
705.0	1.5760	0.4187	41.87	705	1.5760	0.8968	89.68
730.0	1.7527	0.4656	46.56	730	1.7527	0.9974	99.74
755.0	1.7425	0.4629	46.29	755	1.7425	0.9916	99.16
780.0	1.6738	0.4447	44.47	780	1.6738	0.9525	95.25
805.0	1.8391	0.4886	48.86	805	1.8391	1.0465	104.65
830.0	2.0174	0.5359	53.59	830	2.0174	1.1480	114.80
855.0	2.3372	0.6209	62.09	855	2.3372	1.3300	133.00
880.0	2.3663	0.6286	62.86	880	2.3663	1.3466	134.66
905.0	2.7318	0.7257	72.57	905	2.7318	1.5545	155.45
930.0	3.0924	0.8215	82.15	930	3.0924	1.7597	175.97
955.0	3.3165	0.8811	88.11	955	3.3165	1.8873	188.73
980.0	3.5382	0.9400	94.00	980	3.5382	2.0134	201.34

Gold Coated Mirror

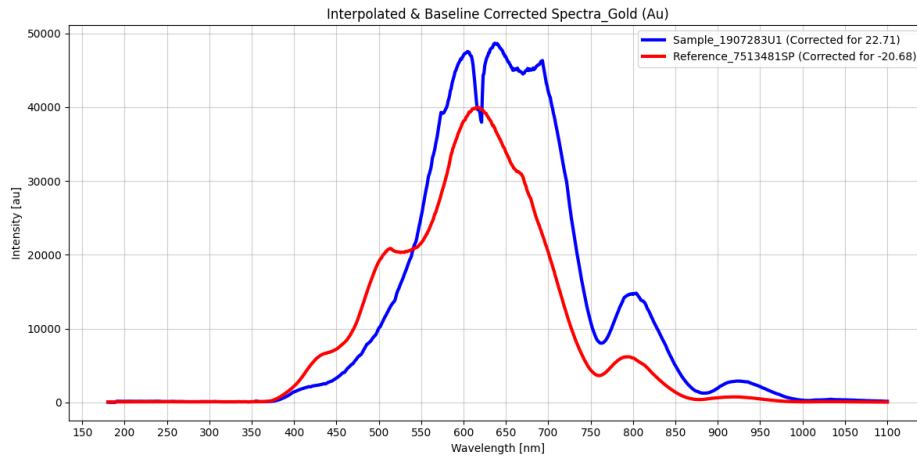


Figure 34: Relative intensity distribution of gold sample and reference

As shown in the gold-coated mirror data presented in figure (34), the reference spectrum is qualitatively similar to that of the aluminum reflectivity measurement only differing primarily in absolute photon intensity with a peak at 620 nm (40,000 au). While the sample's intensity distribution (approx. 48,000 au) largely mirrors the aluminum sample, a significant divergence occurs between 635 nm and 690 nm and by a sharp drop to 37,000 au at 620 nm followed by a steady increase starting at 670 nm.

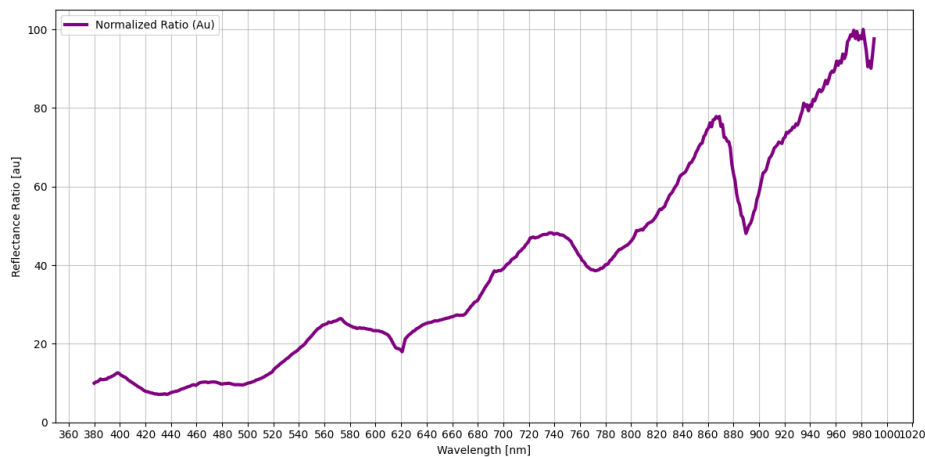


Figure 35: Reflectance ratio of Gold across different wavelengths

The plot of reflectivity for gold coated mirror is too low compared to aluminum in the visible part, starting with a reflectivity 10.26% (380 nm) and gradual increase afterwards upto the end of visible part (700 nm) with a decrease between 429 nm to 453 nm (7.40% and

9.34%). The reflectivity performance increases from 700 nm onwards and showed a peak reflectance 97.82% at 970 nm which is more than twice than the visible red region. However, the behaviour notably changed when normalizing it with respect to 790 nm (perfect spectral shape). the reflectivity is most remarkable also in end of UV region (380 nm, almost 73%) with significant rise in the visible region. The response of the IR regions is almost similar to that normalized respect to maximum ratio.

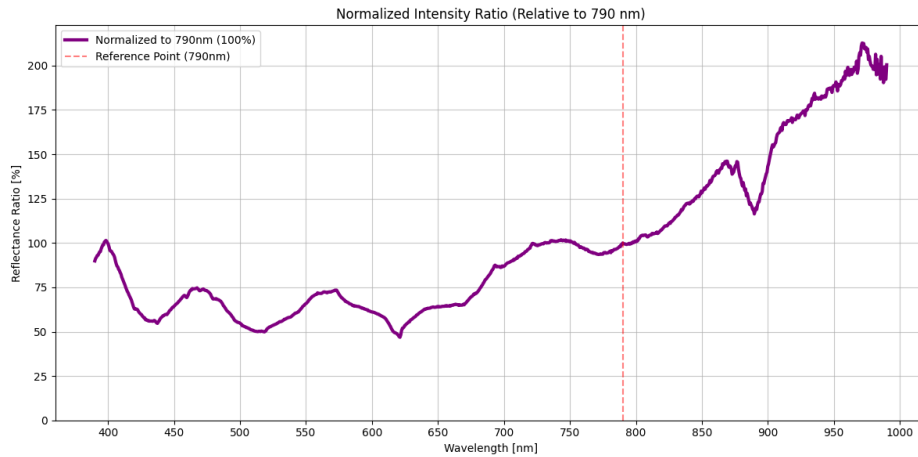


Figure 36: Reflectance ratio of gold across different wavelengths relative to 790 nm

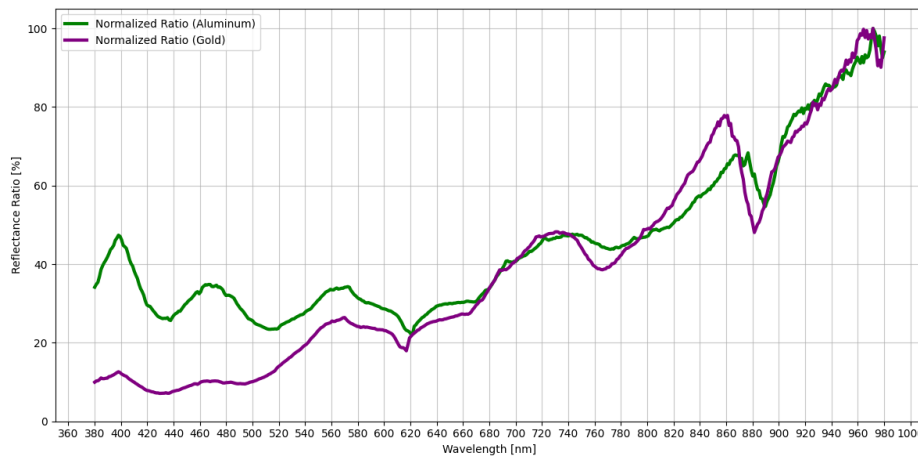


Figure 37: Combined reflectance ratio of Aluminum and Gold

The combined reflectance plot (figure-37) summarize that the reflectivity of Aluminum mirror is strong enough contrasting to gold mirror upto 570 nm, After 620 nm onwards, the reflectivity climbed steadily with an exception for gold at 760 nm (drop). For Aluminum, the reflectivity is almost 70% (880 nm) while it is more than 77% (860 nm) for gold which is 10% more than Aluminum. So it can be concluded that Aluminum is highly reflective at visible region

Table 5: Reflectance Ratio of Gold

Relative to Maximum Ratio				Relative to 790 nm			
Wave-length [nm]	Raw Ratio	Norm. Ratio	Norm. [%]	Wave-length [nm]	Raw Ratio	Norm. Ratio	Norm. [%]
380.0	0.5279	0.1026	10.26	380.0	1.2852	0.7296	72.96
404.6	0.6032	0.1172	11.72	405.0	1.6189	0.9190	91.90
429.2	0.3808	0.0740	7.40	430.0	0.9933	0.5639	56.39
453.8	0.4809	0.0934	9.34	455.0	1.2001	0.6813	68.13
478.3	0.5213	0.1013	10.13	480.0	1.2050	0.6841	68.41
502.9	0.5380	0.1045	10.45	505.0	0.9209	0.5228	52.28
527.5	0.8127	0.1579	15.79	530.0	0.9803	0.5565	55.65
552.1	1.2076	0.2346	23.46	555.0	1.2303	0.6985	69.85
576.7	1.3344	0.2592	25.92	580.0	1.1788	0.6692	66.92
601.3	1.2340	0.2397	23.97	605.0	1.0548	0.5988	59.88
625.8	1.1752	0.2283	22.83	630.0	1.0011	0.5684	56.84
650.4	1.3796	0.2680	26.80	655.0	1.1340	0.6438	64.38
675.0	1.5732	0.3056	30.56	680.0	1.2728	0.7226	72.26
699.6	2.0649	0.4012	40.12	705.0	1.5760	0.8947	89.47
724.2	2.4962	0.4849	48.49	730.0	1.7527	0.9950	99.50
748.8	2.4916	0.4841	48.41	755.0	1.7425	0.9892	98.92
773.3	2.0519	0.3986	39.86	780.0	1.6738	0.9502	95.02
797.9	2.4117	0.4685	46.85	805.0	1.8391	1.0441	104.41
822.5	2.8742	0.5584	55.84	830.0	2.0174	1.1453	114.53
847.1	3.5071	0.6813	68.13	855.0	2.3372	1.3269	132.69
871.7	3.9880	0.7748	77.48	880.0	2.3663	1.3434	134.34
896.3	2.8616	0.5559	55.59	905.0	2.7318	1.5509	155.09
920.8	3.8522	0.7484	74.84	930.0	3.0924	1.7556	175.56
945.4	4.4408	0.8627	86.27	955.0	3.3165	1.8828	188.28
970.0	5.0350	0.9782	97.82	980.0	3.5382	2.0087	200.87

starting blue to red through IR region while gold is poor reflector at UV compared to visible regions but both mirrors system perfect reflector at IR regions.

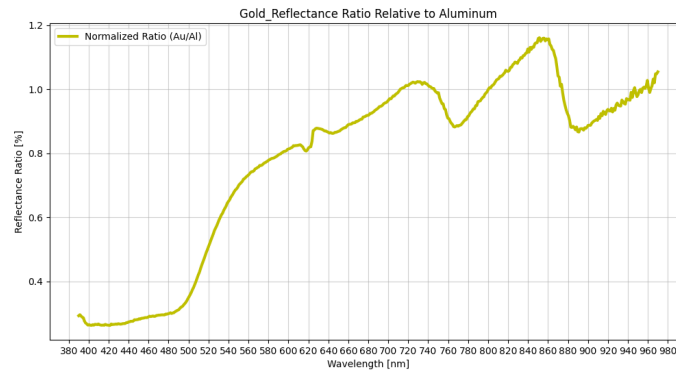


Figure 38: Reflectance of Gold relative to Aluminum

Plot (39) and (38) summarize the relative reflective power of aluminum respect to gold and gold respect to aluminum. So it can be outlined that gold is highly reflective at IR.

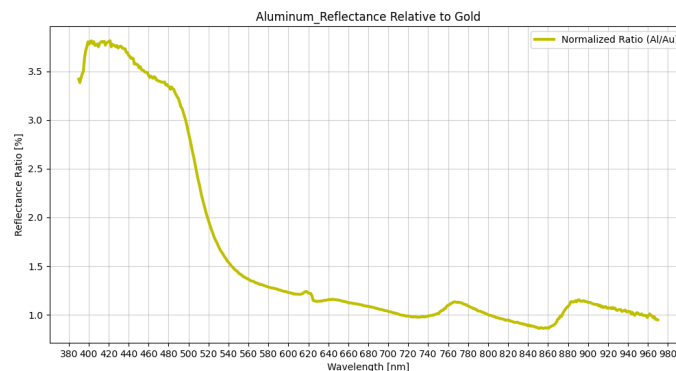


Figure 39: Reflectance of Aluminum relative to Gold

9.2.3 Lab 3 (18.03.2026, light source: discharge-lamp, sample: Al, Ag, Au, dielectric)

The consecutive last (3rd) experiment has been performed by following the same principle described in lab 1 and 2 with an exception with the light source (broadband light source from discharge lamp rather than continuum from non-linear crystal) but same spectral properties. The following spectra represented in figure (40) depicts the relative intensity mapping for various samples (e.g., Al, Au, Ag, Dielectric) along with reference spectrum while figure (41) express intensity profile of different samples at same light source.

Baseline corrected and interpolated

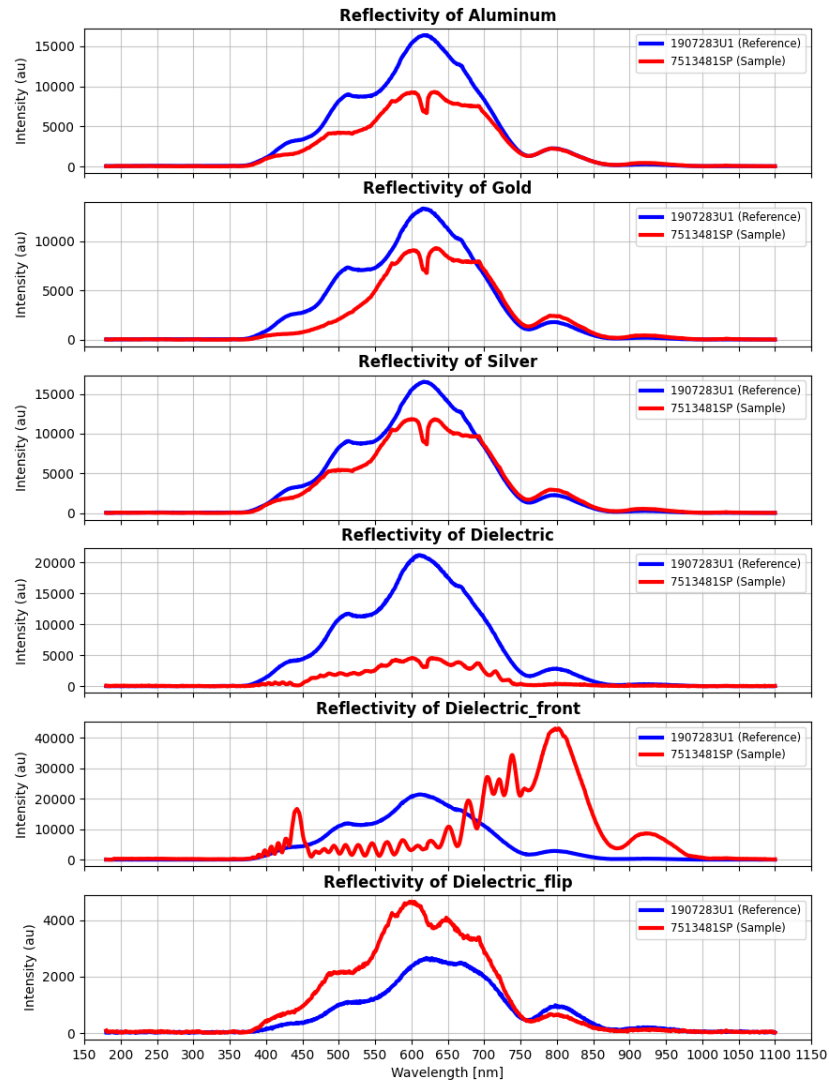


Figure 40: Relative intensity distribution of various sample (Al, Au, Ag, Dielectric) along with reference

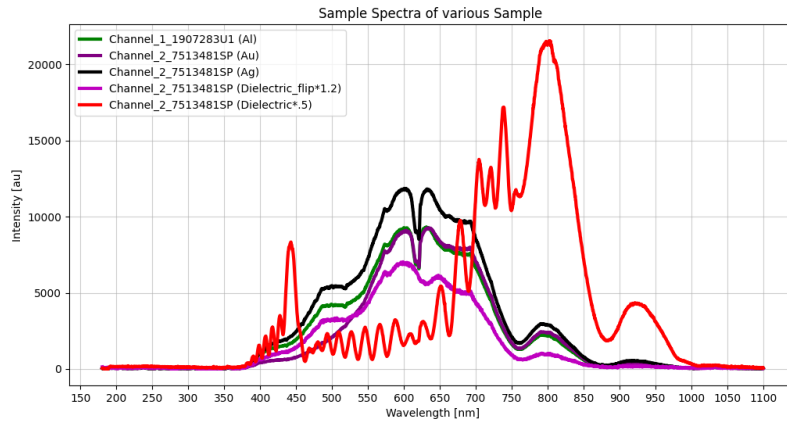


Figure 41: Relative Intensity Distribution of Various Sample at Broadband Light

The following figures (42), (43) and (44) recaps the reflectivity profile for Al, Au and Ag sample. The reflectivity trend is almost same both for Aluminum and Silver at continuum from the discharge lamp while having exception for Gold in UV to Vis regions. The reflectivity data for above mentioned sample is tabulated in the succeeding table (6), (7) and (8) respectively.

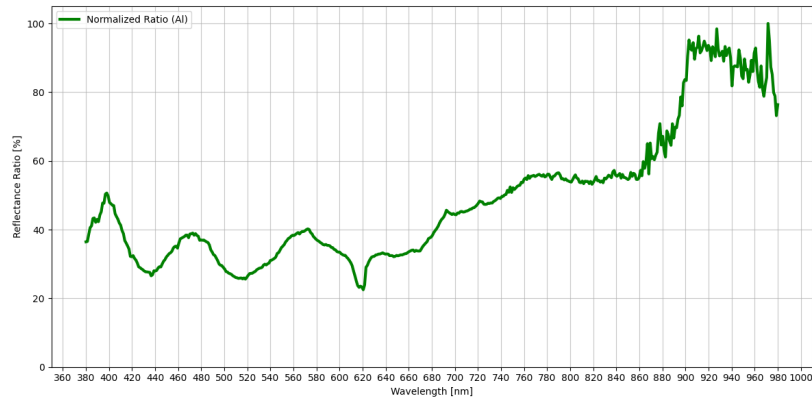


Figure 42: Reflectance Ratios of Aluminum (Al)

Table 6: Reflectance Data for Aluminum

Wavelength [nm]	Raw Ratio	Normalized Ratio	Normalized Ratio [%]
380.0	0.6578	0.3645	36.45
405.0	0.8221	0.4556	45.56
430.0	0.4988	0.2764	27.64
455.0	0.6158	0.3413	34.13
480.0	0.6668	0.3696	36.96
505.0	0.4900	0.2715	27.15
530.0	0.5159	0.2859	28.59
555.0	0.6674	0.3699	36.99
580.0	0.6649	0.3685	36.85
605.0	0.5827	0.3229	32.29
630.0	0.5815	0.3223	32.23
655.0	0.5881	0.3259	32.59
680.0	0.6860	0.3802	38.02
705.0	0.8132	0.4507	45.07
730.0	0.8579	0.4754	47.54
755.0	0.9609	0.5325	53.25
780.0	1.0073	0.5582	55.82
805.0	0.9908	0.5491	54.91
830.0	0.9817	0.5441	54.41
855.0	1.0117	0.5607	56.07
880.0	1.2020	0.6661	66.61
905.0	1.6489	0.9138	91.38
930.0	1.6563	0.9179	91.79
955.0	1.4565	0.8072	80.72
980.0	1.3781	0.7637	76.37

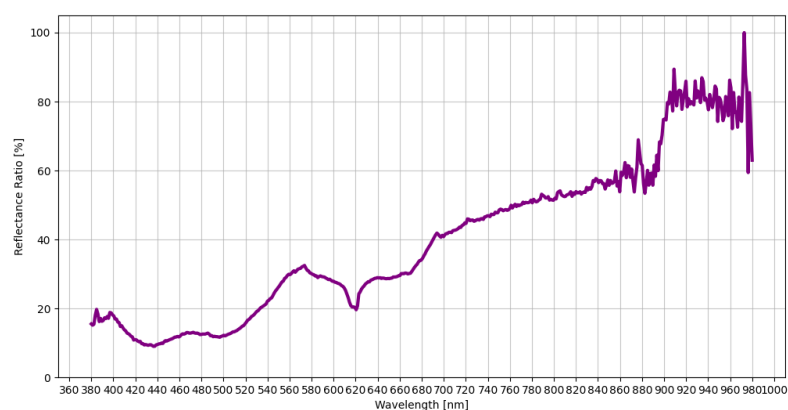


Figure 43: Reflectance Ratios of Gold (Au)

Table 7: Reflectanced Data of Gold (Au)

Wavelength [nm]	Raw Ratio	Normalized Ratio	Normalized Ratio [%]
380.0	0.4036	0.1549	15.49
405.0	0.4168	0.1599	15.99
430.0	0.2434	0.0934	9.34
455.0	0.2982	0.1144	11.44
480.0	0.3259	0.1250	12.50
505.0	0.3277	0.1257	12.57
530.0	0.4983	0.1912	19.12
555.0	0.7422	0.2848	28.48
580.0	0.7832	0.3005	30.05
605.0	0.7061	0.2709	27.09
630.0	0.7136	0.2738	27.38
655.0	0.7565	0.2902	29.02
680.0	0.8894	0.3412	34.12
705.0	1.0894	0.4180	41.80
730.0	1.1929	0.4577	45.77
755.0	1.2706	0.4875	48.75
780.0	1.3257	0.5086	50.86
805.0	1.3807	0.5298	52.98
830.0	1.4508	0.5567	55.67
855.0	1.4804	0.5680	56.80
880.0	1.6184	0.6210	62.10
905.0	2.1180	0.8126	81.26
930.0	2.1390	0.8207	82.07
955.0	1.9006	0.7292	72.92
980.0	1.6416	0.6299	62.99

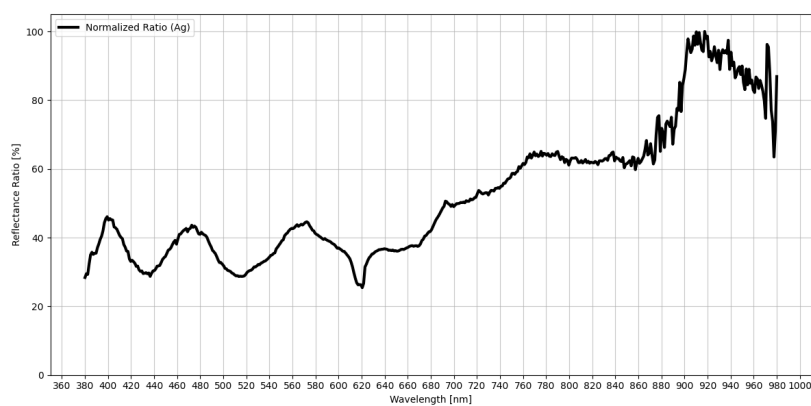


Figure 44: Reflectance Ratios Spectrum of Silver

Table 8: Reflectance Data for Silver

Wavelength [nm]	Raw Ratio	Normalized Ratio	Normalized Ratio [%]
380.0	0.5875	0.2839	28.39
405.0	0.9033	0.4365	43.65
430.0	0.6168	0.2981	29.81
455.0	0.7772	0.3756	37.56
480.0	0.8490	0.4103	41.03
505.0	0.6287	0.3038	30.38
530.0	0.6681	0.3228	32.28
555.0	0.8534	0.4124	41.24
580.0	0.8502	0.4108	41.08
605.0	0.7415	0.3583	35.83
630.0	0.7311	0.3533	35.33
655.0	0.7558	0.3652	36.52
680.0	0.8651	0.4180	41.80
705.0	1.0333	0.4993	49.93
730.0	1.0851	0.5244	52.44
755.0	1.2203	0.5897	58.97
780.0	1.3488	0.6518	65.18
805.0	1.3351	0.6452	64.52
830.0	1.2732	0.6152	61.52
855.0	1.3020	0.6292	62.92
880.0	1.5109	0.7301	73.01
905.0	2.0352	0.9835	98.35
930.0	1.9531	0.9438	94.38
955.0	1.7185	0.8304	83.04
980.0	1.7978	0.8688	86.88

Dielectric Mirror

The subsequent figures (45) and (46) summarize the ratio of the dielectric mirror back and front spot with its associated reflectivity data in table (9) while figure (47) represents its ratio(back_spot/front_spot) with corresponding tabulated data in table (10). In addition, figure (48) represents same dielectric mirror's reflectivity at flipped condition with data in table (11).

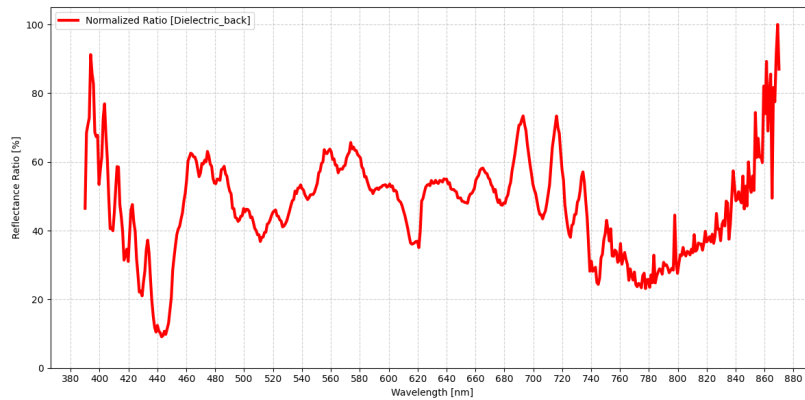


Figure 45: Reflectance Ratios of Dielectric Mirror (back)

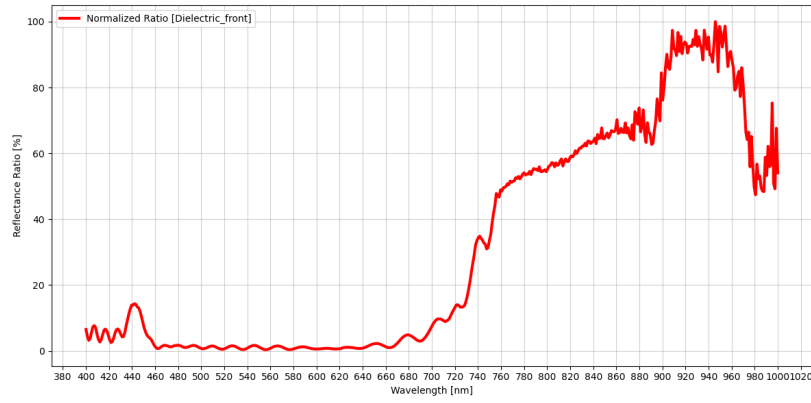


Figure 46: Reflectance Ratios of Dielectric Mirror (front)

Table 9: Dielectric Mirror Reflectivity Data Comparison for Both Spots

Wave [nm]	Raw	Norm	Norm [%]	Wave [nm]	Raw	Norm	Norm [%]
390	0.1944	0.4648	46.48	390	2.0188	0.0773	7.73
410	0.1763	0.4216	42.16	420	1.0623	0.0407	4.07
430	0.0959	0.2292	22.92	450	2.0829	0.0797	7.97
450	0.0921	0.2201	22.01	480	0.4726	0.0181	1.81
470	0.2382	0.5694	56.94	510	0.4078	0.0156	1.56
490	0.2156	0.5153	51.53	540	0.2366	0.0091	0.91
510	0.1625	0.3885	38.85	570	0.3194	0.0122	1.22
530	0.1805	0.4315	43.15	600	0.1533	0.0059	0.59
550	0.2202	0.5265	52.65	630	0.2810	0.0108	1.08
570	0.2472	0.5909	59.09	660	0.3455	0.0132	1.32
590	0.2161	0.5166	51.66	690	0.8039	0.0308	3.08
610	0.1939	0.4636	46.36	720	3.6168	0.1384	13.84
630	0.2248	0.5374	53.74	750	9.1114	0.3488	34.88
650	0.2054	0.4910	49.10	780	14.6741	0.5617	56.17
670	0.2285	0.5463	54.63	810	15.3557	0.5878	58.78
690	0.2948	0.7048	70.48	840	17.2757	0.6613	66.13
710	0.2055	0.4913	49.13	870	18.0931	0.6926	69.26
730	0.1948	0.4658	46.58	900	21.8809	0.8376	83.76
750	0.1793	0.4286	42.86	930	25.4521	0.9743	97.43
770	0.1176	0.2813	28.13	960	26.1248	1.0000	100.00
790	0.1303	0.3114	31.14	990	12.5211	0.4793	47.93
810	0.1410	0.3370	33.70	—	—	—	—
830	0.1679	0.4014	40.14	—	—	—	—
850	0.2339	0.5592	55.92	—	—	—	—
870	0.3638	0.8698	86.98	—	—	—	—
890	0.2602	0.6220	62.20	—	—	—	—
910	0.3626	0.8668	86.68	—	—	—	—
930	0.4149	0.9920	99.20	—	—	—	—

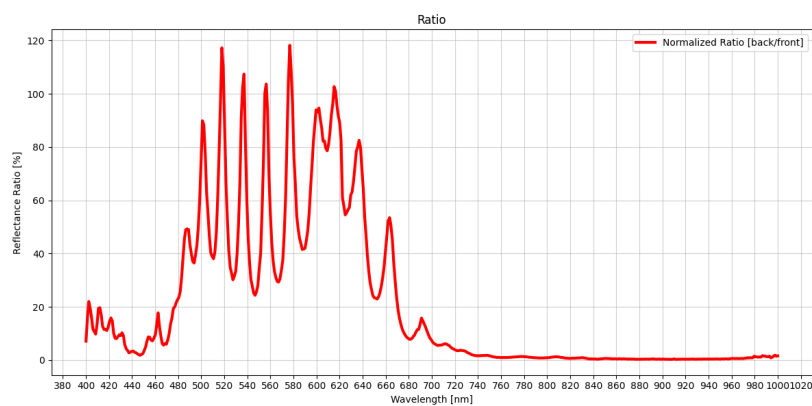


Figure 47: Combined Reflectance Ratios of Dielectric Mirror (back and front)

Table 10: Average Reflectance Data Combing Both Spots

Wavelength [nm]	Raw Ratio	Normalized Ratio	Normalized Ratio [%]
400	1.7703	0.0675	6.75
430	1.4051	0.0536	5.36
460	0.3529	0.0135	1.35
490	0.3798	0.0145	1.45
520	0.1725	0.0066	0.66
550	0.3262	0.0124	1.24
580	0.1557	0.0059	0.59
610	0.2123	0.0081	0.81
640	0.2332	0.0089	0.89
670	0.6398	0.0244	2.44
700	2.0328	0.0775	7.75
730	4.2094	0.1606	16.06
760	13.0412	0.4974	49.74
790	15.2713	0.5825	58.25
820	15.7574	0.6011	60.11
850	17.9610	0.6851	68.51
880	18.9994	0.7247	72.47
910	25.1131	0.9579	95.79
940	26.2164	1.0000	100.00
970	21.8895	0.8350	83.50

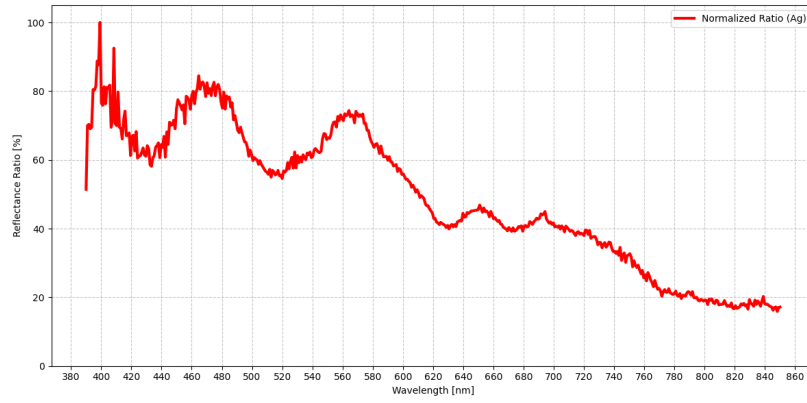


Figure 48: Reflectance Ratios of Dielectric Mirror (flipped)

Table 11: Reflectance Data of Dielectric(Flipped)

Wavelength [nm]	Raw Ratio	Normalized Ratio	Normalized Ratio [%]
390.00	1.8151	0.5144	51.44
409.17	2.4903	0.7057	70.57
428.33	2.3320	0.6609	66.09
447.50	2.4737	0.7010	70.10
466.67	2.9274	0.8296	82.96
485.83	2.6524	0.7517	75.17
505.00	2.1141	0.5991	59.91
524.17	2.0532	0.5818	58.18
543.33	2.2335	0.6329	63.29
562.50	2.5804	0.7313	73.13
581.67	2.2743	0.6445	64.45
600.83	1.9552	0.5541	55.41
620.00	1.5655	0.4436	44.36
639.17	1.5130	0.4288	42.88
658.33	1.5845	0.4490	44.90
677.50	1.4374	0.4073	40.73
696.67	1.4663	0.4155	41.55
715.83	1.3796	0.3910	39.10
735.00	1.2330	0.3494	34.94
754.17	1.0380	0.2942	29.42
773.33	0.8045	0.2280	22.80
792.50	0.7313	0.2073	20.73
811.67	0.6363	0.1803	18.03
830.83	0.6480	0.1836	18.36
850.00	0.6057	0.1716	17.16

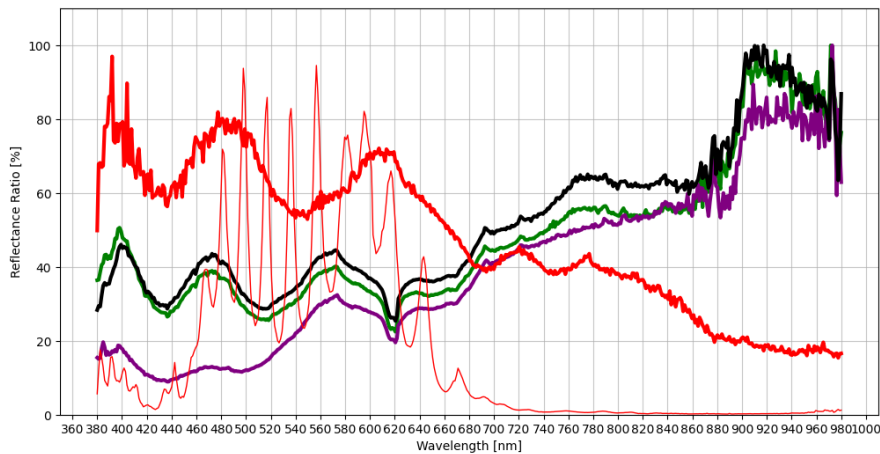


Figure 49: Combined Representation of Reflectance Ratios for Different Sample

The above plot (49) represents the combine picture of all the sample and its respective reflectivity while plot (50) and (51) represents the relative reflectivity of all other samples compared to Aluminum and Gold. It's a bit strange but concrete that the relative reflectivity of other samples compared to Aluminum and Gold is almost same for discharge light source which has indexed in table (12) with a bit different in its decimal points. Mentioning that the integration time was different while taking data measurement for dielectric mirror which also been taken into consideration while plotting the respective data.

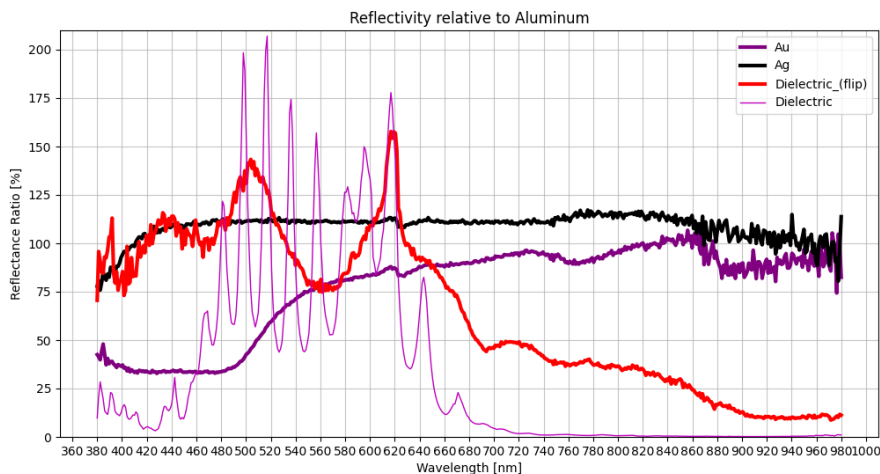


Figure 50: Reflectance Ratios for Different Sample Relative to Aluminum

Table 12: Reflectance Data of Relative to Other Samples relative to Aluminum and Gold

Wavelength [nm]	Raw Ratio	Normalized Ratio	Ratio [%]
380.00	0.4248	0.4248	42.48
404.05	0.3411	0.3411	34.11
429.30	0.3360	0.3360	33.60
454.55	0.3378	0.3378	33.78
479.80	0.3358	0.3358	33.58
503.85	0.4548	0.4548	45.48
529.10	0.6544	0.6544	65.44
554.35	0.7572	0.7572	75.72
579.60	0.8113	0.8113	81.13
604.85	0.8362	0.8362	83.62
628.90	0.8418	0.8418	84.18
654.15	0.8919	0.8919	89.19
679.40	0.8976	0.8976	89.76
704.65	0.9304	0.9304	93.04
729.90	0.9598	0.9598	95.98
753.95	0.9192	0.9192	91.92
779.20	0.9254	0.9254	92.54
804.45	0.9635	0.9635	96.35
829.70	1.0021	1.0021	100.21
854.95	1.0100	1.0100	101.00
879.00	0.9594	0.9594	95.94
904.25	0.8554	0.8554	85.54
929.50	0.8949	0.8949	89.49
954.75	0.9141	0.9141	91.41
980.00	0.8247	0.8247	82.47

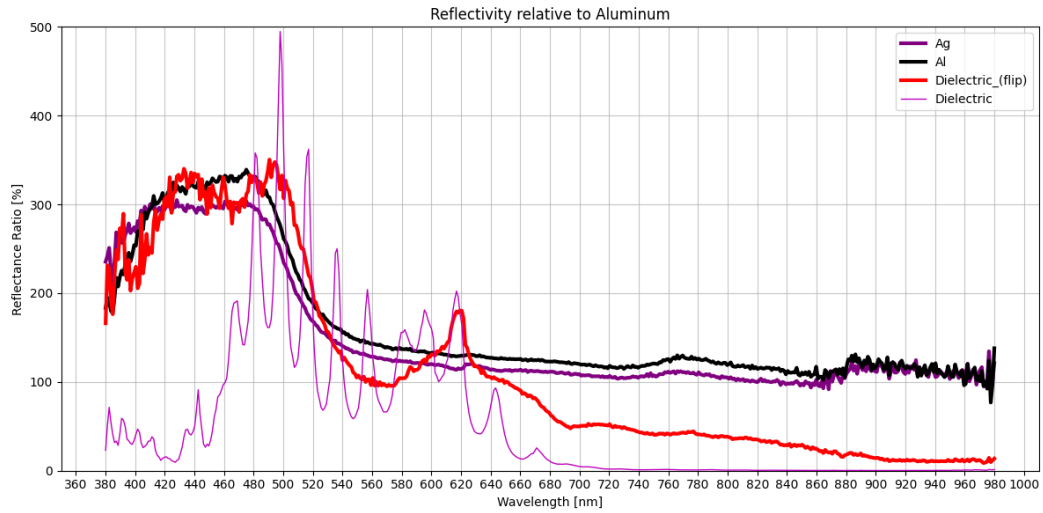


Figure 51: Reflectance Ratios for Different Sample Relative to Gold

9.2.4 Lab 4 (..... light source: fs LASER driven continuum)

10 Discussion

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